

Empirical Evaluation of Security Scanners for Docker Containers

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Acronyms

Acronyms	Definition
CI/CD	Continuous Integration / Continuous Deployment
CISA	Cybersecurity and Infrastructure Security Agency
CLI	Command Line Interface
CVE	Common Vulnerabilities and Exposures
CVSS	Common Vulnerability Scoring System
DB	Database
DevOps	Development and Operations
DevSecOps	Development, Security, and Operations
EOSL	End of Service Life
FIRST	Forum of Incident Response and Security Teams
GQM	Goal-Question-Metric
ISEP	Instituto Superior de Engenharia do Porto
JSON	JavaScript Object Notation
LABCIB	Laboratório de Cibersegurança
NVD	National Vulnerability Database
OASIS	Organization for the Advancement of Structured Information Standards
RCE	Remote Code Execution
SaaS	Software as a Service
SARIF	Static Analysis Results Interchange Format
SAST	Static Application Security Testing
SBOM	Software Bill of Materials
SCA	Software Composition Analysis

Abstract

Container security in DevSecOps environments relies heavily on automated vulnerability scanning tools. However, the effectiveness and consistency of these tools remain insufficiently validated, particularly in complex scenarios involving legacy systems, manually compiled software, and heterogeneous technology stacks. This study presents an empirical evaluation of four vulnerability analysis tools for Docker containers, Trivy, Snyk, Gype, and Docker Scout, using the Goal-Question-Metric (GQM) paradigm as a methodological framework.

The evaluation was conducted across ten Docker images representing distinct risk profiles, including control baselines (alpine:latest, hello-world:latest), End-of-Service-Life (EOSL) systems (Debian stretch-slim), and images containing critical vulnerabilities such as Shellshock (CVE-2014-6271), SambaCry (CVE-2017-7494), Log4Shell (CVE-2021-44228), Spring4Shell (CVE-2022-22965), and Confluence RCE (CVE-2022-26134). Ten metrics organized into four research questions assessed detection effectiveness, precision and noise quality, risk assessment consistency, and operational performance.

The results empirically validate three critical dimensions identified in the theoretical framework. First, the "Structural Blindness" hypothesis was confirmed with a 0% detection rate for CVE-2014-6271 (Shellshock) across all four tools, demonstrating that manually compiled software remains invisible to metadata-based scanning approaches. Second, the "Technical Lag" dimension was validated through Docker Scout's inconsistent behavior in EOSL systems, reporting zero vulnerabilities in Debian 8/9/10 while correctly identifying 848 CVEs in Ubuntu 16.04. Third, the "Risk Assessment Ambiguity" was quantified with a 6.5x variation in vulnerability counts for the same image (vulhub/solr:8.11.0), ranging from 140 (Docker Scout) to 918 (Gype).

Performance analysis revealed significant architectural differences: Trivy achieved the fastest average execution time (4.39s), while Snyk exhibited the highest latency (19.81s) due to its cloud-native architecture. All tools demonstrated high precision in negative controls (0 false positives in hello-world:latest) and unanimous detection of well-documented Java vulnerabilities (Log4Shell, Spring4Shell). The study concludes with use-case-specific recommendations and highlights the need for complementary security strategies beyond automated scanning to address the identified blind spots.

Keywords: Docker security, vulnerability scanning, SAST, SCA, container security, DevSecOps, empirical evaluation, GQM, Trivy, Snyk, Gype, Docker Scout

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1. Introduction and Objectives

The **objective** of this project is to conduct a comparative empirical evaluation between four static application security testing (SAST) tools for containers: Trivy (Aqua Security), Grype (Anchore), Snyk, and Docker Scout. The study aims to measure the effectiveness of these tools in controlled scenarios, contrasting vulnerability detection across different layers (Operating System vs. Application) and installation methods (Package Manager vs. Source Code), answering questions about precision, coverage, and consistency.

Vulnerability analysis in container images constitutes a fundamental pillar of security in DevSecOps environments (Shu et al., 2017). The proliferation of Software Composition Analysis (SCA) tools introduces, however, a problem of ambiguity in risk assessment, where different scanners may report divergent results for the same image (Alfadel et al., 2021).

1.1 Evaluated Tools

Tool	Type	Version	Architecture
Trivy	Open-source (Aqua Security)	0.68.2	Local + Remote DB
Snyk	Commercial (SaaS)	1.1301.2	Cloud-native
Grype	Open-source (Anchore)	0.103.0	Local + Remote DB
Docker Scout	Commercial (Docker Inc.)	1.18.3	SBOM-based + Cloud

Tabela 1- Evaluated Tools

1.2 Tool Selection Justification

The tool selection was based on the study by Imtiaz et al. (2021), which identified Trivy and Grype as representatives of the state of the art in open-source scanners, complemented by Snyk (commercial leader in SCA) and Docker Scout (native solution in the Docker ecosystem).

2. GQM Model for Scanner Evaluation

This study adopts the Goal-Question-Metric (GQM) paradigm proposed by Basili et al. (1994) as a methodological framework to structure the empirical evaluation.

2.1 Goal

Analyze vulnerability scanners in Docker images (Trivy, Gype, Snyk, Docker Scout) with the purpose of evaluating their effectiveness and consistency from the **point of view of a security/DevOps engineer** in the context of a cybersecurity laboratory and production environments.

2.2 Questions and Metrics

Question 1: Detection Effectiveness (Coverage)

Addresses the "Fallacy of Total Coverage" dimension (Section 3.2)

Question: What is the capability of each tool to detect known vulnerabilities in different layers and installation types?

Metric	Description	Type
M1	True Positive Rate - % of injected CVEs correctly identified	Quantitative
M2	False Negatives - Number of critical CVEs in Ground Truth ignored	Quantitative
M3	Detection by Layer - Identifies if vuln is in base image or COPY/RUN instruction	Binary

Tabela 2 - Metrics Question 1

Question 2: Precision and Noise Quality

Addresses the "Hidden Debt of Technical Lag" dimension (Section 3.1)

Question: How reliable are the generated alerts (avoiding alert fatigue)?

Metric	Description	Type
M4	False Positive Rate - Vulns reported in clean images	Quantitative
M5	Version Precision - Identifies backported fixes	Binary

Tabela 3 - Metrics Question 2

Question 3: Risk Assessment Consistency

Addresses the "Ambiguity and Inconsistency" dimension (Section 3.3)

Question: Is there consensus among tools regarding the severity of detected threats?

Metric	Description	Type
M6	Severity Divergence - % of CVEs with different classification among tools	Quantitative
M7	NVD Consonance - Proximity to official database	Qualitative

Tabela 4 - Metrics Question 3

Question 4: Performance and Integration

Relevant for practical CI/CD scenarios

Question: What is the operational impact of using each tool?

Metric	Description	Type
M8	Execution Time - Seconds from start to report	Quantitative
M9	Database Size - Disk space occupied	Quantitative
M10	Output Quality - Supported formats (JSON, SARIF, Table)	Categorical

Tabela 5 - Metrics Question 4

3. Problem Interpretation and Framework

Cloud-Native infrastructure security often rests on the premise that scanning automation ensures container integrity. However, this research challenges this notion, identifying that the security problem lies not only in the existence of software vulnerabilities, but in the **structural inability of auditing tools to detect them effectively in complex scenarios**.

The problem interpretation is structured into **three critical dimensions** that motivate this empirical study:

3.1 The Hidden Debt of "Technical Lag" (Security Inertia)

Container immutability ("Build once, run anywhere"), while advantageous for operational stability, generates a side effect of **silent obsolescence**. As explored in specialized literature, this phenomenon, called *Technical Lag* (Zerouali et al., 2019), describes the temporal discrepancy between production software and the latest security fixes.

The problem is aggravated by operational resistance to updating stable base images, creating a **security technical debt** that accumulates in the lower layers of the infrastructure. Frequently, this debt remains invisible to development teams, who focus only on the application layer, resulting in functionally correct but structurally vulnerable containers.

Link to GQM: This dimension is evaluated through **Question 2 (Precision and Quality)**, specifically by metric **M5 (Version Precision)** which measures the ability of scanners to identify backported fixes in EOSL systems.

3.2 The Fallacy of Total Coverage (Structural Blindness of SAST)

The most critical technological limitation identified in this context lies in the **architectural dependency of scanning tools** (SCA/SAST). Our analysis suggests that these solutions operate predominantly through reading metadata and package manager manifests, rather than inspecting the actual binary code.

This approach creates a **systemic "blind spot"**: any software component introduced into the container outside standard channels, for example, via manual compilation of source code or direct installation scripts, leaves no trace in the system inventory. Consequently, this software becomes **invisible to automated auditing**, generating False Negatives that validate the passage of insecure artifacts to production environments (Plate et al., 2015).

Link to GQM: This dimension is evaluated through **Question 1 (Detection Effectiveness)**, specifically by metrics **M1 (True Positive Rate)**, **M2 (False Negatives)**, and **M3 (Detection by Layer)**.

3.3 Ambiguity and Inconsistency in Risk Assessment

Finally, the security problem is exacerbated by the **absence of universal normalization in severity classification**. The existence of multiple sources of truth (NVD, Vendor Security Advisories, GitHub) introduces significant operational uncertainty (Williams & Dabirsiaghi, 2012).

Different tools use distinct metrics to evaluate the same CVE, varying between **theoretical risk** (based on maximum potential impact) and **contextual risk** (based on the real probability of exploitation in a specific system). This discrepancy results in contradictory evaluations for the same artifact, undermining trust in automation and forcing constant human intervention for result triage, which nullifies the efficiency promised by DevSecOps practices.

Link to GQM: This dimension is evaluated through **Question 3 (Risk Assessment Consistency)**, specifically by metrics **M6 (Severity Divergence)** and **M7 (NVD Consonance)**.

4. Methodology and Dataset

4.1 Dataset Selection and Justification

Vulnerability selection followed a hybrid approach to ensure historical relevance and currency, divided into three categories:

Historical Benchmarks (Regression): Based on the study by Tunde-Onadele et al. (2019), the following were selected:

- Shellshock (CVE-2014-6271)
- SambaCry (CVE-2017-7494)

Modern Threats (Supply Chain): Based on current threat reports (CISA, 2022):

- Log4Shell (CVE-2021-44228)
- Spring4Shell (CVE-2022-22965)
- Atlassian Confluence RCE (CVE-2022-26134)
- Apache Path Traversal (CVE-2021-41773)

Control Scenarios: For false positive validation:

- Alpine Linux latest
- Hello-world latest
- Debian stretch-slim

4.2 Selected Docker Images

#	Image	Target CVE	Category
1	alpine:latest	Control (clean)	Baseline
2	hello-world:latest	Control (minimal)	Baseline
3	debian:stretch-slim	Technical Lag	EOSL
4	vulhub/bash:4.3.0	CVE-2014-6271	Manual Compilation
5	vulhub/httpd:2.4.49	CVE-2021-41773	Application
6	vulhub/spring-webmvc:5.3.17	CVE-2022-22965	Java Library
7	bkimminich/juice-shop:latest	Multiple	Node.js Application
8	vulhub/samba:4.6.3	CVE-2017-7494	EOSL + Library
9	vulhub/solr:8.11.0	CVE-2021-44228	Java Library
10	vulhub/confluence:7.13.6	CVE-2022-26134	Application

Tabela 6 - Docker Images

4.3 Experimental Environment Configuration

To ensure test reproducibility, a local laboratory was configured using Docker and Docker Compose. The experimentation workflow followed these steps:

1. **Environment Instantiation:** Use of *docker-compose up -d* to raise the vulnerable container in isolation (based on Dockerfiles from the Vulhub repository).
2. **Scanner Execution:** Sequential execution of CLI tools directed at the built image:
 - *trivy image --format json*
 - *grype <image> -o json*
 - *snky container test*
 - *docker scout cves --format sarif*
3. **Data Collection:** Redirection of each tool's output to structured files (JSON/SARIF) for subsequent quantitative analysis.

Methodological Note: Analysis of the vulhub/bash:4.3.0 image with the Snyk tool was not possible due to the absence of the Dockerfile in the specified path, constituting an architectural limitation of the tool in scenarios of analyzing pre-built images without build metadata.

4.4 Execution Evidence

Below is the proof of tool execution in the different test scenarios. Each scan generated a structured output file (JSON) for subsequent analysis.

4.4.1 Image 1: *alpine:latest*

Trivy:

time trivy image alpine:latest --format json --output trivy-alpine.json

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/trivy/alpine# time trivy image alpine:latest --format json --output trivy-alpine.json
2025-11-20T18:35:34Z    INFO    [vuln] Vulnerability scanning is enabled
2025-11-20T18:35:34Z    INFO    [secret] Secret scanning is enabled
2025-11-20T18:35:34Z    INFO    [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2025-11-20T18:35:34Z    INFO    [secret] Please see https://trivy.dev/v0.67/docs/scanner/secret#recommendation for faster secret detection
2025-11-20T18:35:34Z    INFO    Detected OS    family="alpine" version="3.22.2"
2025-11-20T18:35:34Z    INFO    [alpine] Detecting vulnerabilities...    os_version="3.22" repository="3.22" pkg_num=16
2025-11-20T18:35:34Z    INFO    Number of language-specific files    num=0

real    0m0.236s
user    0m0.184s
sys     0m0.109s
```

Figure 1- trivy scanner command for alpine image

Grype:

time grype alpine:latest -o json > grype-alpine.json

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/grype/alpine# time grype alpine:latest -o json > grype-alpine.json
✓Loaded image
✓Parsed image
✓Cataloged contents
├── Packages [16 packages]
├── Executables [17 executables]
├── File metadata [81 locations]
├── File digests [81 files]
└── Scanned for vulnerabilities [6 vulnerability matches]
    └── by severity: 0 critical, 0 high, 0 medium, 6 low, 0 negligible
A newer version of grype is available for download: 0.104.0 (installed version is 0.103.0)

real    0m1.696s
user    0m3.236s
sys     0m1.606s
```

Figure 2 - Grype scanner command for alpine image

Snyk:

`time snyk container test alpine:latest --json | tee snyk_alpine_results.json | jq -r "'Scan completed for \\.path). Status: \\.summary) (\\.uniqueCount) unique vulnerabilities found).'"`

```
monteiro_20@LAPTOP-80Q7BQD9:/mnt/c/Users/20joa/OneDrive/Ambiente de Trabalho/lab/lab/conteudo/snyk/alpine$ time snyk container test alpine:latest -
-json | tee snyk_alpine_results.json | jq -r "'Scan completed for \\.path). Status: \\.summary) (\\.uniqueCount) unique vulnerabilities found).'"
Scan completed for alpine:latest. Status: No known vulnerabilities (0 unique vulnerabilities found).

real    0m7.498s
user    0m3.074s
sys     0m0.558s
```

Figure 3 - Snyk scanner command for alpine image

Docker Scout:

`time docker scout cves --format sarif --output scout_alpine.json local://alpine:latest`

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/docker/alpine# time docker scout cves --format sarif --output scout_alpin
e.json Local://alpine:latest
i New version 1.19.0 available (installed version is 1.18.3) at https://github.com/docker/scout-cli
✓ Image stored for indexing
✓ Indexed 20 packages
✓ No vulnerable package detected
✓ Report written to scout_alpine.json

real    0m3.907s
user    0m1.359s
sys     0m0.268s
```

Figure 44 - Docker scanner command for alpine image

4.4.2 Image 2: hello-world:latest

Trivy:

`time trivy image --format json --output trivy_helloworld.json hello-world:latest`

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/trivy/helloworld# time trivy image --format json --output trivy_helloworl
d.json hello-world:latest
2025-12-20T00:28:22Z INFO [vuln] Need to update DB
2025-12-20T00:28:22Z INFO [vuln] Downloading vulnerability DB...
2025-12-20T00:28:22Z INFO [vuln] Downloading artifact... repo="mirror.gcr.io/aquasec/trivy-db:2"
78.71 MiB / 78.71 MiB [-----] 100.00% 8.71 MiB p/s 9.2s
2025-12-20T00:28:34Z INFO [vuln] Artifact successfully downloaded repo="mirror.gcr.io/aquasec/trivy-db:2"
2025-12-20T00:28:34Z INFO [vuln] Vulnerability scanning is enabled
2025-12-20T00:28:34Z INFO [secret] Secret scanning is enabled
2025-12-20T00:28:34Z INFO [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2025-12-20T00:28:34Z INFO [secret] Please see https://trivy.dev/v0.67/docs/scanner/secret#recommendation for faster secret detection
2025-12-20T00:28:34Z INFO Number of language-specific files num=0

real    0m11.985s
user    0m1.633s
sys     0m2.335s
```

Figure 5 - trivy scanner command for helloworld image

Grype:

time gype hello-world:latest -o json > gype_helloworld.json

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/grype/helloworld# time gype hello-world:latest -o json > gype_helloworld.json
[updated]
✓ Vulnerability DB
✓ Loaded image
✓ Parsed image
✓ Cataloged contents
  [✓ Packages
  [✓ Executables
✓ Scanned for vulnerabilities
  by severity: 0 critical, 0 high, 0 medium, 0 low, 0 negligible
  by status: 0 fixed, 0 not-fixed, 0 ignored [0000] WARN current database is invalid error=the vulnerability dat
A newer version of gype is available for download: 0.104.2 (installed version is 0.103.0)

real    0m35.330s
user    0m30.379s
sys     0m10.156s
```

Figure 6 - Grype scanner command for helloworld image

Snyk:

*time snyk container test hello-world:latest --json | tee snyk_helloworld_results.json | jq -r
"Scan completed for \(.path)."*

```
monteiro_20@LAPTOP-88QT8QD9:/mnt/c/Users/20joa/OneDrive/Ambiente de Trabalho/lab/lab/conteudo/snyk/helloworld$ time snyk
container test hello-world:latest --json | tee snyk_helloworld_results.json | jq -r "Scan completed for \(.path)."
```

Figure 7 - Snyk scanner command for helloworld image

Docker Scout:

time docker scout cves --format sarif --output scout_helloworld.json local://hello-world:latest

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/docker/helloworld# time docker scout cves --format sarif --output scout_helloworld.json local://hello-world:latest
i New version 1.19.0 available (installed version is 1.18.3) at https://github.com/docker/scout-cli
✓ SBOM of image already cached, 0 packages indexed
✓ 1 exception obtained
✓ No vulnerable package detected
✓ Report written to scout_helloworld.json

real    0m2.280s
user    0m0.190s
sys     0m0.169s
```

Figure 8 – Docker Scout scanner command for helloworld image

4.4.3 Image 3: debian:stretch-slim

Trivy:

*time trivy image --format json --output trivy_debian_slim.json
debian@sha256:f186c98e81a67c2fba071c2a71545c4c7e5696c8fd9a66529939d0af337d5a73*

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/trivy/debian# time trivy image --format json --output trivy_debian_slim.j
son debian@sha256:f186c98e81a67c2fba071c2a71545c4c7e5696c8fd9a66529939d0af337d5a73
2025-12-20T00:52:07Z INFO [vuln] Vulnerability scanning is enabled
2025-12-20T00:52:07Z INFO [secret] Secret scanning is enabled
2025-12-20T00:52:07Z INFO [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanni
ng
2025-12-20T00:52:07Z INFO [secret] Please see https://trivy.dev/docs/v0.68/guide/scanner/secret#recommendation for
faster secret detection
2025-12-20T00:52:08Z INFO Detected OS family="debian" version="9.13"
2025-12-20T00:52:08Z INFO [debian] Detecting vulnerabilities... os_version="9" pkg_num=76
2025-12-20T00:52:08Z INFO Number of language-specific files num=0
2025-12-20T00:52:08Z WARN This OS version is no longer supported by the distribution family="debian" version=
"9.13"
2025-12-20T00:52:08Z WARN The vulnerability detection may be insufficient because security updates are not provide
d
real 0m1.378s
user 0m0.999s
sys 0m0.277s
```

Figure 9 - Trivy scanner command for debian image

Grype:

time grype

debian@sha256:f186c98e81a67c2fba071c2a71545c4c7e5696c8fd9a66529939d0af337d5a73 -
o json > grype_debian_slim.json

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/grype/debian# time grype debian@sha256:f186c98e81a67c2fba071c2a71545c4c7e
5696c8fd9a66529939d0af337d5a73 -o json > grype_debian_slim.json
✓Loaded image          debian@sha256:f186c98e81a67c2fba071c2a71545c4c7e5696c8fd9a66529939d0af337d5a73
✓Parsed image          sha256:1db2bf7e4dc4cf6761fdbdb0afdaf4198c0608d6de9d15a66204ff3fef9ae7b9
a0726c53276e546c1776cad4fc9f83e42789e20838a9d475f3b4ac0c693946d8
✓Cataloged contents
├── Packages            [76 packages]
├── Executables          [684 executables]
├── File metadata        [2,713 locations]
├── File digests         [2,713 files]
✓Scanned for vulnerabilities [209 vulnerability matches]
├── by severity: 23 critical, 46 high, 20 medium, 46 low, 74 negligible
A newer version of grype is available for download: 0.104.2 (installed version is 0.103.0)
real 0m1.808s
user 0m2.851s
sys 0m0.905s
```

Figure 10 - Grype scanner command for debian image

Snyk:

time snyk container test

debian@sha256:f186c98e81a67c2fba071c2a71545c4c7e5696c8fd9a66529939d0af337d5a73 -
-json | tee snyk_debian_slim_results.json | jq -r '"Scan completed for \(.path)."'

```
monteiro_20@LAPTOP-80QT8QD9:/mnt/c/Users/20joa/OneDrive/Ambiente de Trabalho/Lab/Lab/conteudo/snyk/debian$ time snyk con
tainer test debian@sha256:f186c98e81a67c2fba071c2a71545c4c7e5696c8fd9a66529939d0af337d5a73 --json | tee snyk_debian_slim
_results.json | jq -r '"Scan completed for \(.path)."'
Scan completed for debian@sha256:f186c98e81a67c2fba071c2a71545c4c7e5696c8fd9a66529939d0af337d5a73.
real 0m8.391s
user 0m4.433s
sys 0m0.815s
```

Figure 11 - Snyk scanner command for debian image

Docker Scout:

time docker scout cves --format sarif --output scout_debian_slim.json

local://debian@sha256:f186c98e81a67c2fba071c2a71545c4c7e5696c8fd9a66529939d0af337
d5a73


```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/docker/debian# time docker scout cves --format sarif --output scout_debian_slim.json local://debian@sha256:f186c98e81a67c2fba071c2a71545c4c7e5696c8fd9a66529939d0af337d5a73
i New version 1.19.0 available (installed version is 1.18.3) at https://github.com/docker/scout-cli
✓ Image stored for indexing
✓ Indexed 102 packages
✓ No vulnerable package detected
✓ Report written to scout_debian_slim.json

real    0m6.947s
user    0m4.814s
sys     0m0.757s
```

Figure 12 - Docker Scout scanner command for debian image

4.4.4 Image 4: vulhub/bash:CVE-2014-6271 (Shellshock)

Trivy:

`time trivy image --input bash-vulhub.tar --format json --output trivy.json`

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/vulhub/bash/CVE-2014-6271# time trivy image --input bash-vulhub.tar --format json --output trivy.json
2025-11-20T18:09:55Z INFO [vuln] Vulnerability scanning is enabled
2025-11-20T18:09:55Z INFO [secret] Secret scanning is enabled
2025-11-20T18:09:55Z INFO [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2025-11-20T18:09:55Z INFO [secret] Please see https://trivy.dev/v0.67/docs/scanner/secret#recommendation for faster secret detection
2025-11-20T18:09:55Z INFO Detected OS family="debian" version="8.10"
2025-11-20T18:09:55Z INFO [debian] Detecting vulnerabilities... os_version="8" pkg_num=131
2025-11-20T18:09:55Z INFO Number of language-specific files num=0
2025-11-20T18:09:55Z WARN Using severities from other vendors for some vulnerabilities. Read https://trivy.dev/v0.67/docs/scanner/vulnerability#severity-selection for details.
2025-11-20T18:09:55Z WARN This OS version is no longer supported by the distribution family="debian" version="8.10"
2025-11-20T18:09:55Z WARN The vulnerability detection may be insufficient because security updates are not provided

real    0m0.175s
user    0m0.154s
sys     0m0.127s
```

Figure 13 - Trivy scanner command for bash image

Grype:

`time grype oci-archive:./bash-vulhub.tar -o json > grype.json`

```
monteiro_20@LAPTOP-80QT8QD9:/mnt/c/Users/20joa/OneDrive/Ambiente de Trabalho/lab/lab/vulhub/bash/CVE-2014-6271$ ls
1.png 2.png README.md README.zh-cn.md docker-compose.yml safe.cgi victim.cgi
monteiro_20@LAPTOP-80QT8QD9:/mnt/c/Users/20joa/OneDrive/Ambiente de Trabalho/lab/lab/vulhub/bash/CVE-2014-6271$ time snyk container test vulhub/bash:4.3 --file=Dockerfile --json | tee snyk_bash_results.json | jq -r '"Scan completed for \(.path)."'
Scan completed for vulhub/bash:4.3.

real    0m6.266s
user    0m3.039s
sys     0m0.453s
```

Figure 14 - Grype scanner command for bash image

Snyk:

`time snyk container test vulhub/bash:4.3 --file=Dockerfile --json | tee snyk_bash_results.json | jq -r '"Scan completed for \(.path)."'`

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/vulhub/bash/CVE-2014-6271# ls
1.png 2.png README.md README.zh-cn.md bash-vulhub.tar docker-compose.yml safe.cgi trivy.json victim.cgi
root@DESKTOP-KVVIT6D:/home/vieira/lab/vulhub/bash/CVE-2014-6271# time grype oci-archive:./bash-vulhub.tar -o json > grype.json
✓Parsed image sha256:6bb66831a48e4b19a53ff44baf8426c23569625f7350b1a379bf6b74c74671e8
✓Cataloged contents 23c8f71ce46c5ec07225ed8c85f20a1b1adc44ef6313d6814dc9c0515e043059
├── Packages [132 packages]
├── Executables [1,019 executables]
├── File metadata [7,370 locations]
├── File digests [7,370 files]
└── Scanned for vulnerabilities [468 vulnerability matches]
    └── by severity: 2 critical, 70 high, 162 medium, 97 low, 137 negligible
A newer version of grype is available for download: 0.104.0 (installed version is 0.103.0)

real    0m2.467s
user    0m4.518s
sys     0m2.253s
```

Figure 15 - Snyk scanner command for bash image

NOTE: <aside> ERROR: Dockerfile not found in the specified path. The Snyk tool requires the Dockerfile for analysis of pre-built images. </aside>

Docker Scout:

```
time docker scout cves --format sarif --output scout_bash_shellshock.json
local://vulhub/bash:4.3.0-with-httpd
```

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/vulhub# time docker scout cves --format sarif --output scout_bash_shellshock.json
local://vulhub/bash:4.3.0-with-httpd
  i New version 1.19.0 available (installed version is 1.18.3) at https://github.com/docker/scout-cli
  ✓ SBOM of image already cached, 176 packages indexed
  ✓ No vulnerable package detected
  ✓ Report written to scout_bash_shellshock.json

real    0m1.935s
user    0m0.206s
sys     0m0.151s
```

Figure 16 – Docker Scout scanner command for bash image

4.4.5 Image 5: vulhub/httpd:2.4.49

Trivy:

```
time trivy image --format json --output trivy_httpd_rce.json
vulhub/httpd@sha256:a6c32432b1ed40818488e238769e962a4151ef3b74b54416d9d5d396ba8cab5f
```

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/trivy/httpd# time trivy image --format json --output trivy_httpd_rce.json
vulhub/httpd@sha256:a6c32432b1ed40818488e238769e962a4151ef3b74b54416d9d5d396ba8cab5f
2025-12-20T00:48:04Z      INFO    [vuln] Vulnerability scanning is enabled
2025-12-20T00:48:04Z      INFO    [secret] Secret scanning is enabled
2025-12-20T00:48:04Z      INFO    [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2025-12-20T00:48:04Z      INFO    [secret] Please see https://trivy.dev/docs/v0.68/guide/scanner/secret#recommendation for faster secret detection
2025-12-20T00:48:04Z      INFO    Detected OS      family="debian" version="10.10"
2025-12-20T00:48:04Z      INFO    [debian] Detecting vulnerabilities...  os_version="10" pkg_num=110
2025-12-20T00:48:04Z      INFO    Number of language-specific files  num=0
2025-12-20T00:48:04Z      WARN    Using severities from other vendors for some vulnerabilities. Read https://trivy.dev/docs/v0.68/guide/scanner/vulnerability#severity-selection for details.
2025-12-20T00:48:04Z      WARN    This OS version is no longer supported by the distribution  family="debian" version="10.10"
2025-12-20T00:48:04Z      WARN    The vulnerability detection may be insufficient because security updates are not provided

real    0m0.123s
user    0m0.136s
sys     0m0.032s
```

Figure 17 - Trivy scanner command for httpd image

Grype:

```
time grype
vulhub/httpd@sha256:a6c32432b1ed40818488e238769e962a4151ef3b74b54416d9d5d396ba8cab5f -o json > grype_httpd_rce.json
```

```

root@DESKTOP-KVVIT6D: /home/vieira/lab/conteudo/grype/httpd# time grype vulhub/httpd[sha256:a6c32432b1ed40818488e238769e9
62a4151ef3b74b54416d9d5d396ba8cab5f -o json > grype_httpd_rce.json
  ✓Loaded image                                vulhub/httpd[sha256:a6c32432b1ed40818488e238769e962a4151ef3b74b54416d9d5d396ba8cab5
  ✓Parsed image                                sha256:391841fe52846b3bc9b25166dd022c5a6646f1a0a5d364a51dc45e9209045765
  ✓Cataloged contents                          6947924aa6123fcfc6812fbcb9f968cde69bb699bd9cecae17306cc6cf4a4eafab
    ├── ✓Packages                             [111 packages]
    ├── ✓Executables                           [881 executables]
    ├── ✓File metadata                         [2,857 locations]
    └── ✓File digests                          [2,857 files]
  ✓Scanned for vulnerabilities                 [398 vulnerability matches]
    ├── by severity: 34 critical, 134 high, 113 medium, 29 low, 84 negligible (4 unknown)
A newer version of grype is available for download: 0.104.2 (installed version is 0.103.0)

real    0m4.165s
user    0m4.295s
sys      0m2.840s

```

Figure 18 - Grype scanner command for httpd image

Snyk:

time snyk container test

```
vulhub/httpd@sha256:a6c32432b1ed40818488e238769e962a4151ef3b74b54416d9d5d396ba8cab5f --json | tee snyk_httpd_results.json | jq -r "'Scan completed for %(path)s.'"
```

[illegible]

Figure 19 - Snyk scanner command for httpd image

Docker Scout:

`time docker scout cves --format sarif --output scout_httpd_rce.json`
`local://vulhub/httpd@sha256:a6c32432b1ed40818488e238769e962a4151ef3b74b54416d9d5d396ba8cab5f`

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/docker/httpd# time docker scout cves --format sarif --output scout_httpd_rce.json
1 New version 1.19.0 available (installed version is 1.18.3) at https://github.com/docker/scout-cli
✓ Image stored for indexing
✓ Indexed 163 packages
✓ No vulnerable package detected
✓ Report written to scout_httpd_rce.json

real    0m10.653s
user    0m9.581s
sys     0m1.136s
```

Figure 20 – Docker Scout scanner command for httpd image

4.4.6 Image 6: vulhub/spring-webmvc:5.3.17 (Spring4Shell)

Trivy:

`time trivy image vulhub/spring-webmvc:5.3.17 --format json --output trivy-spring.json`

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/trivy/spring# time trivy image vulhub/spring-webmvc:5.3.17 --format json --output trivy-spring.json
2025-11-20T18:27:42Z INFO [vuln] Vulnerability scanning is enabled
2025-11-20T18:27:42Z INFO [secret] Secret scanning is enabled
2025-11-20T18:27:42Z INFO [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2025-11-20T18:27:42Z INFO [secret] Please see https://trivy.dev/v0.67/docs/scanner/secret#recommendation for faster secret detection
2025-11-20T18:27:42Z INFO Detected OS family="debian" version="11.3"
2025-11-20T18:27:42Z INFO [debian] Detecting vulnerabilities... os_version="11" pkg_num=143
2025-11-20T18:27:42Z INFO Number of language-specific files num=1
2025-11-20T18:27:42Z INFO [jar] Detecting vulnerabilities...
2025-11-20T18:27:42Z WARN Using severities from other vendors for some vulnerabilities. Read https://trivy.dev/v0.67/docs/scanner/vulnerability#severity-selection for details.

real    0m0.298s
user    0m0.230s
sys     0m0.080s
```

Figure 21 - Trivy scanner command for spring image

Grype:

`time grype vulhub/spring-webmvc:5.3.17 -o json > grype-spring.json`

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/grype/spring# time grype vulhub/spring-webmvc:5.3.17 -o json > grype-spring.json
✓Loaded image vulhub/spring-webmvc:5.3.17
✓Parsed image sha256:dceea93a08684d3f2faedde60a6dcfba1c63be8984a3d4e1b19f7e4bbf55d9e
✓Cataloged contents c18780659189652bd7eb98a6890af1f5277e4980cb33e23102965ca06a7777c2
├─✓Packages [190 packages]
├─✓Executables [824 executables]
├─✓File metadata [6,459 locations]
├─✓File digests [6,459 files]
└─✓Scanned for vulnerabilities [512 vulnerability matches]
   └─by severity: 35 critical, 138 high, 175 medium, 53 low, 107 negligible (4 unknown)
A newer version of grype is available for download: 0.104.0 (installed version is 0.103.0)

real    0m6.446s
user    0m6.457s
sys     0m2.784s
```

Figure 22- Grype scanner command for spring image

Snyk:

`time snyk container test vulhub/spring-webmvc:5.3.17 --json | tee snyk_spring_results.json | jq -r "'Scan completed for \u{002F}\u{002F}(\u{002F}).'"`

```
monteiro_20@LAPTOP-89Q18QD9:/mnt/c/Users/20joa/OneDrive/Ambiente de Trabalho/Lab/lab/conteudo/snyk/spring$ time snyk con
tainer test vulhub/spring-webmvc:5.3.17 --json | tee snyk_spring_results.json | jq -r '"Scan completed for \(.path)."'
Scan completed for vulhub/spring-webmvc:5.3.17/spring-webmvc.

real    0m19.361s
user    0m13.436s
sys     0m2.468s
```

Figure 23 - Snyk scanner command for spring image

Docker Scout:

`time docker scout cves --format sarif --output scout_spring4shell.json local://vulhub/spring-webmvc:5.3.17`

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/docker/spring# time docker scout cves --format sarif --output scout_spr
ing4shell.json local://vulhub/spring-webmvc:5.3.17
i New version 1.19.0 available (installed version is 1.18.3) at https://github.com/docker/scout-cli
✓ Image stored for indexing
✓ Indexed 249 packages
X Detected 49 vulnerable packages with a total of 193 vulnerabilities
✓ Report written to scout_spring4shell.json

real    0m18.390s
user    0m21.373s
sys     0m3.130s
```

Figure 24 – Docker Scout scanner command for spring image

4.4.7 Image 7: bkimminich/juice-shop:latest

Trivy:

`time trivy image --format json --output trivy_juiceshop.json bkimminich/juice-shop@sha256:261556674452a4d45ee7a0a326808525292db17a36ecef046a86d48d82ce2f9`

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/trivy/juiceshop# time trivy image --format json --output trivy_juiceshop.
json bkimminich/juice-shop@sha256:261556674452a4d45ee7a0a326808525292db17a36ecef046a86d48d82ce2f9
2025-12-20T00:51:00Z INFO [vuln] Vulnerability scanning is enabled
2025-12-20T00:51:00Z INFO [secret] Secret scanning is enabled
2025-12-20T00:51:00Z INFO [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2025-12-20T00:51:00Z INFO [secret] Please see https://trivy.dev/docs/v0.68/guide/scanner/secret#recommendation for
faster secret detection
2025-12-20T00:51:11Z INFO Detected OS family="debian" version="12.12"
2025-12-20T00:51:11Z INFO [debian] Detecting vulnerabilities... os_version="12" pkg_num=10
2025-12-20T00:51:11Z INFO Number of language-specific files num=1
2025-12-20T00:51:11Z INFO [node-pkg] Detecting vulnerabilities...
2025-12-20T00:51:11Z WARN Using severities from other vendors for some vulnerabilities. Read https://trivy.dev/doc
s/v0.68/guide/scanner/vulnerability#severity-selection for details.

real    0m10.681s
user    0m4.011s
sys     0m1.547s
```

Figure 25 - Trivy scanner command for juiceshop image

Grype:

`time grype bkimminich/juice-shop@sha256:261556674452a4d45ee7a0a326808525292db17a36ecef046a86d48d82ce2f9 -o json > grype_juiceshop.json`


```

root@DESKTOP-KVVIT6D:/home/vieira/lab/contendo/grype/juiceshop# time grype bkimminich/juice-shop@sha256:261556674452a4d4
5ee7a0a326808525292db17a36cefc046a86d48d82ce2f9 -o json > grype_juiceshop.json
✓Loaded image                                     sha256:859475888c3eaab0ab2d32fc0f4ba39736501e11525aecf53ce1d2f1c3c8565de
✓Parsed image                                     fd0415c67691dcacf8127fac95414bb1acd384c972697c9cb4edb7a8fc046d6f8
✓Categorized contents
├── ✓Packages                                     [1,132 packages]
├── ✓Executables                                 [286 executables]
├── ✓File metadata                             [2,386 locations]
├── ✓File digests                               [2,386 files]
✓Scanned for vulnerabilities [62 vulnerability matches]
├── by severity: 8 critical, 19 high, 22 medium, 1 low, 12 negligible
A newer version of grype is available for download: 0.104.2 (installed version is 0.103.0)

real    0m9.779s
user    0m8.953s
sys     0m3.212s

```

Figure 26 - Grype scanner command for juiceshop image

Snyk:

```
time snyk container test bkimminich/juice-
shop@sha256:261556674452a4d45ee7a0a326808525292db17a36ecef046a86d48d82ce2f9 --
json | tee snyk_juiceshop_results.json | jq -r "'Scan completed for \\(.path).'"
```

[illegible]

Figure 27 - Snyk scanner command for juiceshop image

Docker Scout:

```
time docker scout cves --format sarif --output scout_juiceshop.json local://bkimminich/juice-shop@sha256:261556674452a4d45ee7a0a326808525292db17a36ecef046a86d48d82ce2f9
```

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/docker/juiceshop# time docker scout cves --format sarif --output scout_juiceshop.json local://bkimminich/juice-shop@sha256:261556674452a4d45ee7a0a326808525292db17a36eefc046a86d48d82ce2f9
✓ New version 1.19.0 available (installed version is 1.18.3) at https://github.com/docker/scout-cli
✓ Image stored for indexing
✓ Indexed 1002 packages
✗ Detected 29 vulnerable packages with a total of 55 vulnerabilities
✓ Report written to scout_juiceshop.json

real    0m25.283s
user    0m24.397s
sys     0m3.782s
```

Figure 28 - Docker Scout scanner command for juiceshop image

4.4.8 Image 8: vulhub/samba:4.6.3 (SambaCry)

Trivy:

`time trivy image vulhub/samba:4.6.3 --format json --output trivy-samba.json`

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/trivy/samba# time trivy image vulhub/samba:4.6.3 --format json --output trivy-samba.json
on
2025-11-20T18:24:16Z INFO [vuln] Vulnerability scanning is enabled
2025-11-20T18:24:16Z INFO [secret] Secret scanning is enabled
2025-11-20T18:24:16Z INFO [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2025-11-20T18:24:16Z INFO [secret] Please see https://trivy.dev/v0.67/docs/scanner/secret#recommendation for faster secret detection
2025-11-20T18:24:22Z INFO [python] Licenses acquired from one or more METADATA files may be subject to additional terms. Use '-debug' flag to see all affected packages.
2025-11-20T18:24:22Z INFO Detected OS family="ubuntu" version="16.04"
2025-11-20T18:24:22Z INFO [ubuntu] Detecting vulnerabilities... os.version="16.04" pkg_num=324
2025-11-20T18:24:22Z INFO Number of language-specific files num=0
2025-11-20T18:24:23Z WARN This OS version is no longer supported by the distribution family="ubuntu" version="16.04"
2025-11-20T18:24:23Z WARN The vulnerability detection may be insufficient because security updates are not provided

real    0m6.710s
user    0m4.250s
sys     0m2.349s
```

Figure 29 - Trivy scanner command for samba image

Grype:

`time grype vulhub/samba:4.6.3 -o json > grype-samba.json`

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/grype/samba# time grype vulhub/samba:4.6.3 -o json > grype-samba.json
✓ Loaded image vulhub/samba:4.6.3
✓ Parsed image sha256:e37802d9a4e56e467bb3d225b76b28a80561a465c57c622e7f6bcd247f1f7ca5
✓ Cataloged contents 98a437f498089dc6d8a6e0877c2e04f55619979b42a7a6a907996ddf1f12a82d
├── Packages [330 packages]
├── Executables [1,910 executables]
├── File metadata [15,827 locations]
├── File digests [15,827 files]
✓ Scanned for vulnerabilities [3045 vulnerability matches]
├── by severity: 1 critical, 223 high, 5788 medium, 1341 low, 314 negligible
A newer version of grype is available for download: 0.104.0 (installed version is 0.103.0)

real    0m16.857s
user    0m28.374s
sys     0m7.074s
```

Figure 30 - Grype scanner command for samba image

Snyk:

`time snyk container test vulhub/samba:4.6.3 --json | tee snyk_samba_results.json | jq -r '"Scan completed for \(.path)."'`

```
monteiro_20@LAPTOP-88QT8QD9:/mnt/c/Users/20joa/OneDrive/Ambiente de Trabalho/Lab/lab/conteudo/snyk/samba$ time snyk container test vulhub/samba:4.6.3 --json | tee snyk_samba_results.json | jq -r '"Scan completed for \(.path)."'
Scan completed for vulhub/samba:4.6.3/samba.

real    0m22.743s
user    0m15.048s
sys     0m4.697s
```

Figure 31 - Snyk scanner command for samba image

Docker Scout:

*time docker scout cves --format sarif --output scout_samba_cry.json
local://vulhub/samba:4.6.3*

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/docker/samba# time docker scout cves --format sarif --output scout_samba_cry.json
i New version 1.19.0 available (installed version is 1.18.3) at https://github.com/docker/scout-cli
✓ SBOM of image already cached, 414 packages indexed
✗ Detected 52 vulnerable packages with a total of 848 vulnerabilities
✓ Report written to scout_samba_cry.json

real    0m5.703s
user    0m1.466s
sys     0m0.593s
```

Figure 32 - Docker Scout scanner command for samba image

4.4.9 Image 9: vulhub/solr:8.11.0 (Log4Shell)

Trivy:

time trivy image vulhub/solr:8.11.0 --format json --output trivy-solr.json

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/trivy/log4j# time trivy image vulhub/solr:8.11.0 --format json --output trivy-solr.json
2025-11-20T18:16:49Z INFO [vuln] Vulnerability scanning is enabled
2025-11-20T18:16:49Z INFO [secret] Secret scanning is enabled
2025-11-20T18:16:49Z INFO [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2025-11-20T18:16:49Z INFO [secret] Please see https://trivy.dev/v0.67/docs/scanner/secret#recommendation for faster secret detection
2025-11-20T18:16:49Z INFO Detected OS family="debian" version="8.6"
2025-11-20T18:16:49Z INFO [debian] Detecting vulnerabilities... os_version="8" pkg_num=166
2025-11-20T18:16:49Z INFO Number of language-specific files num=1
2025-11-20T18:16:49Z INFO [jar] Detecting vulnerabilities...
2025-11-20T18:16:49Z WARN Using severities from other vendors for some vulnerabilities. Read https://trivy.dev/v0.67/docs/scanner/vulnerability#severity-selection for details.
2025-11-20T18:16:49Z WARN This OS version is no longer supported by the distribution family="debian" version="8.6"
2025-11-20T18:16:49Z WARN The vulnerability detection may be insufficient because security updates are not provided

real    0m0.659s
user    0m0.314s
sys     0m0.122s
```

Figure 33 - Trivy scanner command for log4j image

Grype:

time grype vulhub/solr:8.11.0 -o json > grype-log4j.json

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/grype/log4j# time grype vulhub/solr:8.11.0 -o json > grype-solr.json
✓Loaded image
✓Parsed image sha256:052794134d434bc2db0775211589beb372412af333a262dbf7977d28ec8d2142
✓Cataloged contents b62888f0cda62c3fff21f0d680f227a2244c82e4cb46d25d6066f571841e64ba
├── Packages [521 packages]
├── Executables [953 executables]
├── File metadata [7,246 locations]
├── File digests [7,246 files]
└── Scanned for vulnerabilities [918 vulnerability matches]
    ├── by severity: 32 critical, 211 high, 421 medium, 124 low, 130 negligible
    └── A newer version of grype is available for download: 0.104.0 (installed version is 0.103.0)

real    0m10.331s
user    0m15.740s
sys     0m6.390s
```

Figure 34 - Grype scanner command for log4j image

Snyk:

time snyk container test vulhub/solr:8.11.0 --json | tee snyk_log4j_results.json | jq -r '"Scan completed for \(.path)."'

```
monteiro_20@LAPTOP-80Q18QD9:/mnt/c/Users/20joa/OneDrive/Ambiente de Trabalho/Lab/Lab/conteudo/snyk/log4j$ time snyk container test vulhub/solr:8.11.0 --json | tee snyk_log4j_results.json | jq -r '"Scan completed for \(.path)."'
Scan completed for vulhub/solr:8.11.0/solr.

real    0m26.908s
user    0m24.620s
sys     0m5.390s
```

Figure 35 - Snyk scanner command for log4j image

Docker Scout:

*time docker scout cves --format sarif --output scout_solr_log4shell.json
local://vulhub/solr:8.11.0*

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/vulhub/log4j/CVE-2021-44228# time docker scout cves --format sarif --output scout_solr_log4shell.json local://vulhub/solr:8.11.0
i New version 1.19.0 available (installed version is 1.18.3) at https://github.com/docker/scout-cli
✓ SBOM of image already cached, 537 packages indexed
X Detected 51 vulnerable packages with a total of 140 vulnerabilities
✓ Report written to scout_solr_log4shell.json

real    0m2.253s
user    0m0.277s
sys     0m0.177s
```

Figure 36 – Docker Scout scanner command for log4j image

4.4.10 Image 10: vulhub/confluence:7.13.6 (Confluence RCE)

Trivy:

time trivy image vulhub/confluence:7.13.6 --format json --output trivy-confluence.json

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/trivy/confluence# time trivy image vulhub/confluence:7.13.6 --format json --output trivy-confluence.json
2025-11-20T18:30:53Z INFO [vuln] Vulnerability scanning is enabled
2025-11-20T18:30:53Z INFO [secret] Secret scanning is enabled
2025-11-20T18:30:53Z INFO [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2025-11-20T18:30:53Z INFO [secret] Please see https://trivy.dev/v0.67/docs/scanner/secret#recommendation for faster secret detection
2025-11-20T18:31:04Z INFO Detected OS family="ubuntu" version="20.04"
2025-11-20T18:31:04Z INFO [ubuntu] Detecting vulnerabilities... os_version="20.04" pkg_num=147
2025-11-20T18:31:04Z INFO Number of language-specific files num=1
2025-11-20T18:31:04Z INFO [jar] Detecting vulnerabilities...
2025-11-20T18:31:04Z WARN This OS version is no longer supported by the distribution family="ubuntu" version="20.04"
2025-11-20T18:31:04Z WARN The vulnerability detection may be insufficient because security updates are not provided

real    0m11.641s
user    0m0.676s
sys     0m3.137s
```

Figure 37 - Trivy scanner command for confluence image

Grype:

time grype vulhub/confluence:7.13.6 -o json > grype-confluence.json

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/conteudo/grype/confluence# time grype vulhub/confluence:7.13.6 -o json > grype-confluence.json
✓Loaded image
✓Parsed image sha256:cfbc60f8430d4d9ab16dce64fc03614f5cd2cedca34ef584be751347a6f05d0f139827ddfe4e619168d533dd41c6394fa23ef3de9e2e19c597c77820109390cd
✓Cataloged contents
├── ✓Packages [1,764 packages]
├── ✓Executables [902 executables]
├── ✓File metadata [6,415 locations]
├── ✓File digests [6,415 files]
✓Scanned for vulnerabilities [889 vulnerability matches]
├── by severity: 24 critical, 219 high, 467 medium, 155 low, 24 negligible
└── A newer version of grype is available for download: 0.104.0 (installed version is 0.103.0)

real    0m19.678s
user    0m42.871s
sys     0m14.536s
```

Figure 38 - Grype scanner command for confluence image

Snyk:

time snyk container test vulhub/confluence:7.13.6 --json | tee snyk_confluence_results.json | jq -r "'Scan completed for ((.path))'"

```
monteiro_20@LAPTOP-88QT8QD9:/mnt/c/Users/20joa/OneDrive/Ambiente de Trabalho/lab/lab/conteudo/snyk/confluence$ time snyk
container test vulhub/confluence:7.13.6 --json | tee snyk_confluence_results.json | jq -r '"Scan completed for \(.path)
"'
Scan completed for vulhub/confluence:7.13.6/confluence.

real    1m0.564s
user    0m54.899s
sys     0m17.453s
```

Figure 39 - Snyk scanner command for confluence image

Docker Scout:

time docker scout cves --format sarif --output scout_confluence.json
local://vulhub/confluence:7.13.6

```
root@DESKTOP-KVVIT6D:/home/vieira/lab/vulhub/confluence/CVE-2022-26134# time docker scout cves --format sarif --output s
cout_confluence.json local://vulhub/confluence:7.13.6
i New version 1.19.0 available (installed version is 1.18.3) at https://github.com/docker/scout-cli
✓ SBOM of image already cached, 1581 packages indexed
X Detected 154 vulnerable packages with a total of 367 vulnerabilities
✓ Report written to scout_confluence.json

real    0m2.849s
user    0m0.469s
sys     0m0.170s
```

Figure 40 - Docker Scout scanner command for confluence image

5. Tool Analysis

5.1 Docker Scout - Empirical Analysis

5.1.1 Technical Scanner Information

The following table summarizes the technical specifications of the Docker Scout scanner used in this evaluation.

Parameter	Value
Scanner	Docker Scout
Version	1.18.3
Manufacturer	Docker Inc.
Output Format	JSON
Method	SBOM-based vulnerability scanning
Database	Cloud-based (Docker Hub Advisory Database)
Type	Cloud-native, integrated into Docker CLI

Tabela 7 - Technical Information Docker Scout

5.1.2. Consolidated Results

This section presents the aggregated results across all ten Docker images analysed during the evaluation. The table below shows the number of packages scanned, execution time, and vulnerability distribution by severity level.

#	Image	Packages	Time (s)	Total	CRIT	HIGH	MED	LOW
1	alpine:latest	20	3.9	0	0	0	0	0
2	hello-world:latest	0	2.3	0	0	0	0	0
3	debian:stretch-slim	102	6.9	0	0	0	0	0
4	bash (Shellshock)	176	1.9	0	0	0	0	0
5	httpd:2.4.49	163	10.7	0	0	0	0	0
6	juice-shop:latest	1002	25.3	55	8	18	22	1
7	solr:8.11.0 (Log4Shell)	537	2.3	140	31	54	47	8
8	samba:4.6.3	414	5.7	848	~10	~80	~400	~350
9	spring-webmvc:5.3.17	249	18.4	193	~15	~50	~80	~48
10	confluence:7.13.6	1581	2.8	367	20	93	199	52

Tabela 8 - Results Docker Images

Summary Statistics: Total Vulnerabilities Detected: 1,603 | Average Scan Time: ~8.2 seconds

5.1.3. Target CVE Detection (Ground Truth)

The ground truth analysis evaluates Docker Scout's ability to detect specific, known critical vulnerabilities that were deliberately included in the test images.

CVE	Name	Image	Detected	CVSS	Observation
CVE-2014-6271	Shellshock	bash	NO	10.0	Manually compiled Bash
CVE-2017-7494	SambaCry	samba	NO	10.0	Gap in CVE-package mapping

CVE-2021-44228	Log4Shell	solr	YES	10.0	log4j-core@2.14.1
CVE-2021-45046	Log4Shell Variant	solr	YES	9.0	log4j-core@2.14.1
CVE-2022-22965	Spring4Shell	spring-webmvc	YES	9.8-10.0	Multiple Spring components
CVE-2022-26134	Confluence RCE	confluence	NO	9.8	Application vulnerability (OGNL)
CVE-2021-41773	Apache Path Traversal	httpd	NO	7.5	Application vulnerability

Tabela 9 - Docker Scout CVE Detection

Detection Metrics

- **Detection Rate:** 3/7 target CVEs (42.8%)
- **True Positives:** 3 (Log4Shell, Log4Shell Variant, Spring4Shell)
- **False Negatives:** 4 (Shellshock, SambaCry, Confluence RCE, Apache Path Traversal)
- **False Positives:** 0 (clean control images)

5.1.4 Detailed Analysis by Category

Control Images (Baseline)

The control images (alpine:latest and hello-world:latest) correctly returned zero vulnerabilities, confirming the absence of false positives in minimalist, updated images. Alpine contained 20 packages and was scanned in 3.9 seconds, while hello-world, being a minimal image without an operating system, contained 0 packages.

End of Support Life (EOSL) Systems

A critical finding emerged regarding EOSL systems. Debian-based images (versions 8, 9, and 10) returned empty JSON results despite indexing packages, while Ubuntu 16.04 was fully analysed and returned 848 vulnerabilities. This inconsistent behaviour represents a significant limitation in the scanner's coverage of legacy systems.

Shellshock Analysis (CVE-2014-6271)

The bash image running on Debian 8.10 indexed 176 packages but returned zero vulnerabilities. The expected Shellshock vulnerability was not detected because Bash was manually compiled rather than installed via the package manager, demonstrating a fundamental limitation of SBOM-based scanning approaches.

Successfully Detected Vulnerabilities

Log4Shell Detection (CVE-2021-44228):

The Solr 8.11.0 image demonstrated successful detection of the Log4Shell vulnerability. Docker Scout identified 140 total vulnerabilities including 31 critical severity issues. The scanner correctly identified [log4j-core@2.14.1](#) as the vulnerable component with CVSS 10.0, and recommended upgrading to version 2.15.0 for remediation.

Spring4Shell Detection (CVE-2022-22965):

The spring-webmvc:5.3.17 image analysis returned 193 vulnerabilities with Spring4Shell correctly identified across multiple components including spring-webmvc, spring-beans, and spring-core. The scanner provided accurate CVSS scores (9.8-10.0) and recommended version 5.3.18 as the fix.

SambaCry Analysis (Critical Discovery):

The samba:4.6.3 image running Ubuntu 16.04 presented a significant finding. While Docker Scout detected 848 vulnerabilities in the underlying system packages (including critical issues in glibc, systemd, and the Linux kernel), it failed to detect CVE-2017-7494 (SambaCry) with CVSS 10.0. This suggests a gap in the CVE-to-Samba package mapping within the vulnerability database.

5.1.5 Critical Discoveries

Inconsistent Behaviour in EOSL Systems

Docker Scout exhibits differentiated behaviour between old Linux distributions. Debian-based images (versions 8, 9, and 10) return empty JSON results with rules: [] in JSON format, while Ubuntu 16.04 receives complete analysis with 848 reported vulnerabilities. This inconsistency represents a specific limitation with old Debian support rather than a general EOSL system problem.

Structural Blindness (Confirmed)

The Shellshock case confirms the structural blindness hypothesis. Software installed outside package managers through manual compilation (wget, configure, make install) remains invisible to SBOM-based scanners. This represents an inherent architectural limitation rather than a deficiency specific to Docker Scout.

Application-Level CVE Limitations

Application logic-level CVEs are systematically undetected. Both the Confluence RCE (CVE-2022-26134 via OGNL injection) and Apache Path Traversal (CVE-2021-41773/42013) were not detected because SBOM-based scanners analyse dependencies rather than code logic. This category of vulnerability requires complementary DAST or IAST tools.

5.1.6 GQM Metrics (Goal-Question-Metric)

Question 1: Detection Effectiveness (M1, M2, M3)

Metric	Value	Observation
M1 - True Positive Rate	42.8% (3/7)	Log4Shell and Spring4Shell successfully detected
M2 - False Negatives	4 CVEs	Shellshock, SambaCry, Confluence RCE, Apache Path Traversal
M3 - Detection by Layer	Yes	Identifies exact package and fixed version

Tabela 10 - Results Question 1 Docker Scout

Question 2: Precision and Noise Quality (M4, M5)

Metric	Value	Observation
M4 - False Positive Rate	0% (0/2)	Alpine and hello-world without false positives
M5 - Version Precision	Limited	Does not detect backports in EOSL systems (Debian)

Tabela 11 - Results Question 2 Docker Scout

Question 3: Risk Assessment Consistency (M6, M7)

CVSS scores are reported directly from authoritative sources (NVD/Vendor) demonstrating high consonance with industry standards. Additional contextual information includes EPSS scores, specific fixed versions, and links to security advisories.

Question 4: Performance and Integration (M8, M9, M10)

Metric	Value	Observation
M8 - Execution Time	Average: 8.2s	Varies between 1.9s (bash) and 25.3s (juice-shop)
M9 - Database Size	Cloud-based	No local disk occupation
M10 - Output Quality	Excellent	Standardised SARIF 2.1.0, easy parsing

Tabela 12 - Results Question 3 Docker Scout

5.1.7. Identified Strengths

1. Excellent Log4Shell detection (CVE-2021-44228) with accurate CVSS 10.0 severity assessment
2. Comprehensive Spring4Shell detection (CVE-2022-22965) across multiple Spring components
3. Zero false positives in clean control images, demonstrating reliable baseline behaviour
4. Effective analysis of Java/Maven and Node.js dependency ecosystems
5. Detailed vulnerability information including CVSS scores, EPSS predictions, and remediation guidance
6. Native Docker CLI integration facilitating seamless CI/CD pipeline adoption

5.1.8 Identified Limitations

1. Inconsistent behaviour across EOSL systems: Debian 8/9/10 return empty JSON while Ubuntu 16.04 is fully analysed
2. Shellshock not detected due to manual compilation outside package manager scope
3. SambaCry not detected suggesting gaps in CVE-to-package mapping for certain software
4. Application-level CVEs (Confluence RCE, Apache Path Traversal) require complementary DAST/IAST tools

5.1.9. Supplementary Data

Vulnerability Distribution by Severity

Severity	Count	Percentage
CRITICAL	~70	4.4%
HIGH	~313	19.5%

MEDIUM	~730	45.5%
LOW	~490	30.6%
TOTAL	1,603	100%

Tabela 13 - Vulnerability Distribution

Performance Analysis

Scan time does not scale linearly with package count. Lightweight images (0-200 packages) complete in 1.9-6.9 seconds, medium images (200-600 packages) in 2.3-10.7 seconds, and heavy images (>1000 packages) benefit from optimisations completing in 2.8-25.3 seconds.

5.2 Trivy - Empirical Analysis

5.2.1 Technical Scanner Information

The following table summarizes the technical specifications of the Trivy scanner used in this evaluation.

Parameter	Value
Scanner	Trivy
Manufacturer	Aqua Security
Method	Package Metadata + Dependency Analysis
Output Format	JSON
Database	Local (automatically downloaded)
Type	Open-source, local executable
Supported Languages	OS packages, Java (JAR), Node.js, Python, Ruby, Go, Rust, etc.

Tabela 14 - Technical Information Trivy

5.2.2. Consolidated Results

This section presents the aggregated results across all ten Docker images analysed during the evaluation. The table includes operating system details, End of Support Life (EOSL) status, package counts, and vulnerability distribution by severity level.

#	Image	OS	EOSL	Pkgs	Total	CRIT	HIGH	MED	LOW	Time
1	alpine:latest	Alpine 3.22.2	No	16	0	0	0	0	0	0.24s
2	hello-world	N/A	N/A	0	0	0	0	0	0	12.0s*
3	debian:stretch-slim	Debian 9.13	Yes	76	68	2	28	4	34	1.38s
4	bash (Shellshock)	Debian 8.10	Yes	131	258	20	90	77	52	0.18s
5	httpd:2.4.49	Debian 10.10	Yes	110	260	23	100	106	22	0.12s
6	spring-webmvc	Debian 11.3	No	187	480	36	117	174	147	0.30s
7	juice-shop	Debian 12.12	No	1133	67	8	22	22	15	10.7s

8	samba:4.6.3	Ubuntu 16.04	Yes	324	1579	1	68	987	523	6.71s
9	solr (Log4Shell)	Debian 8.6	Yes	510	747	121	264	258	77	0.66s
10	confluence	Ubuntu 20.04	Yes	1730	804	27	218	441	118	11.6s

Tabela 15 - Results Trivy

Summary Statistics: Total Vulnerabilities Detected: 4,263 | Average Scan Time: ~4.4 seconds

*Note: * hello-world time includes initial vulnerability database download.*

5.2.3. Target CVE Detection (Ground Truth)

The ground truth analysis evaluates Trivy's ability to detect specific, known critical vulnerabilities that were deliberately included in the test images.

CVE	Name	Image	Detected	Package	CVSS	Observation
CVE-2014-6271	Shellshock	bash	NO	-	10.0	Manually compiled
CVE-2017-7494	SambaCry	samba	NO	-	10.0	Not mapped
CVE-2021-44228	Log4Shell	solr	YES	log4j-core	10.0	JAR dependency
CVE-2022-22965	Spring4Shell	spring-webmvc	YES	spring-beans	9.8	JAR dependency
CVE-2022-22965	Spring4Shell	confluence	YES	spring-beans	9.8	JAR dependency
CVE-2022-26134	Confluence RCE	confluence	NO	-	9.8	Application vuln
CVE-2021-41773	Apache Path Trav.	httpd	NO	-	7.5	Application vuln
CVE-2021-42013	Apache Path Trav.	httpd	NO	-	9.8	Application vuln

Tabela 16 - CVE Detection Trivy

Detection Metrics

- **Detection Rate:** 3/7 target CVEs (42.8%)
- **True Positives:** 3 (Log4Shell, Spring4Shell in 2 images)
- **False Negatives:** 4 (Shellshock, SambaCry, Confluence RCE, Apache Path Traversal)
- **False Positives:** 0 (clean control images)

5.2.4. Detailed Analysis by Category

Control Images (Baseline)

The control images correctly returned zero vulnerabilities, confirming the absence of false positives. Alpine Linux 3.22.2 (8.21 MB, 16 packages) completed scanning in 0.236 seconds, while hello-world (0.01 MB, no packages) required 11.985 seconds due to the initial vulnerability database download. Both results demonstrate correct baseline behaviour.

End of Support Life (EOSL) Systems

Debian Stretch-Slim Analysis:

The debian:stretch-slim image (Debian 9.13, EOSL since 2022) with 76 packages returned 68 vulnerabilities including 2 critical and 28 high severity issues. The majority lack available fixes, demonstrating the Technical Lag problem inherent to outdated images. Scan completed in 1.378 seconds.

Shellshock Analysis (CVE-2014-6271) - Critical False Negative:

The bash image running Debian 8.10 (Jessie EOSL) contained 131 packages and returned 258 vulnerabilities (20 critical, 90 high). However, the expected Shellshock vulnerability (CVE-2014-6271, CVSS 10.0) was not detected despite the image containing vulnerable Bash 4.3.

Root Cause - Structural Blindness: Bash was manually compiled from source code using wget, configure, make, and make install rather than installed via the package manager. Trivy exclusively analyses packages installed via package managers (apt/dpkg), rendering manually compiled software invisible to the scanner. Of the detected vulnerabilities, 219 have available fixes while 39 remain affected without remediation options.

Apache HTTP Server Analysis (CVE-2021-41773/42013):

The httpd:2.4.49 image on Debian 10.10 (Buster EOSL) returned 260 vulnerabilities (23 critical, 100 high) across 110 packages in 0.123 seconds. The expected Apache Path Traversal vulnerabilities were not detected as these represent application/configuration level vulnerabilities rather than dependency issues. Fix status indicates 168 vulnerabilities have remediations available while 92 remain affected.

Successfully Detected Vulnerabilities

Spring4Shell Detection (CVE-2022-22965):

The spring-webmvc:5.3.17 image demonstrated successful Java dependency detection. Running on Debian 11.3 (Bullseye) with 187 packages (143 OS + 44 JAR), the scanner identified 480 vulnerabilities including 36 critical severity issues in 0.298 seconds. Trivy correctly detected Spring4Shell through JAR file analysis, identifying vulnerable packages org.springframework:spring-beans and org.springframework:spring-webmvc at version 5.3.17 with CVSS 9.8, recommending upgrade to version 5.3.18.

OWASP Juice Shop Analysis:

The juice-shop image on Debian 12.12 (Bookworm, current) contained 1133 packages (10 OS + 1123 Node.js) and returned 67 vulnerabilities (8 critical, 22 high) in 10.681 seconds. As an intentionally vulnerable application for security training, the majority of vulnerabilities reside in Node.js dependencies. Top vulnerable packages include jsonwebtoken (4 CVEs), express-jwt (2 CVEs), and crypto-js (1 critical CVE). This demonstrates Trivy's capability to analyse package.json and node_modules ecosystems.

Log4Shell Detection (CVE-2021-44228) - Critical Success:

The solr:8.11.0 image demonstrated successful detection of Log4Shell, one of the most critical vulnerabilities in recent years. Running on Debian 8.6 (Jessie EOSL) with 510 packages (166 OS + 344 JAR), the scanner identified 747 vulnerabilities including 121 critical severity issues in 0.659 seconds. Trivy correctly identified org.apache.logging.log4j:log4j-core version 2.14.1 as vulnerable with CVSS 10.0, recommending upgrade to version 2.17.0. The high critical count reflects both Log4Shell and numerous vulnerabilities in old Java dependencies combined with the outdated Debian 8 base system.

Confluence Analysis:

The confluence:7.13.6 image on Ubuntu 20.04 (expired LTS) contained 1730 packages (147 OS + 1583 JAR) and returned 804 vulnerabilities (27 critical, 218 high) in 11.641 seconds. CVE-2022-26134 (Confluence RCE) was not detected as this represents an application-level Confluence vulnerability not present in dependencies. However, CVE-2022-22965 (Spring4Shell) was successfully detected in the included Spring Framework libraries.

SambaCry Analysis - Discovery of False Negative:

The samba:4.6.3 image on Ubuntu 16.04 (Xenial, EOSL since 2021) presented the highest vulnerability count at 1579 (1 critical, 68 high, 987 medium, 523 low) across 324 packages in 6.710 seconds. Despite containing vulnerable Samba 4.6.3, CVE-2017-7494 (SambaCry, CVSS 10.0) was not explicitly detected.

Possible Causes:

1. Non-standard Samba installation method
2. Incomplete CVE-to-Ubuntu package mapping in Trivy's database
3. Limitation in correlation between package version and CVE identifier

5.2.5. Critical Discoveries

Structural Blindness (Shellshock Case)

The Shellshock case confirms structural blindness as a fundamental limitation. Software installed outside package managers through manual compilation (wget, tar, configure, make install) leaves no trace in the package manager's inventory system.

Trivy depends exclusively on package manager metadata (apt, dpkg, yum, apk), rendering any software installed through alternative channels invisible to detection.

Application-Level CVE Limitations

Application logic-level CVEs are systematically undetected. CVE-2022-26134 (Confluence OGNL Injection), CVE-2021-41773, and CVE-2021-42013 (Apache Path Traversal) all represent vulnerabilities in application logic or configuration rather than dependencies. Dependency-focused Static Application Security Testing (SAST) cannot analyse business logic, execution flows, insecure configurations, or injection points.

Gap in CVE Mapping (SambaCry)

CVE-2017-7494 was not detected despite Samba 4.6.3 being present. This suggests the local vulnerability database may contain incomplete Ubuntu-to-Samba CVE mappings, or the Samba installation method does not conform to expected package manager patterns. This highlights the importance of periodic vulnerability database updates and the potential for mapping gaps.

5.2.6. GQM Metrics (Goal-Question-Metric)

Question 1: Detection Effectiveness (M1, M2, M3)

Metric	Value	Observation
M1 - True Positive Rate	42.8% (3/7)	Log4Shell and Spring4Shell detected
M2 - False Negatives	4 CVEs	Shellshock, SambaCry, Confluence RCE, Apache PT
M3 - Detection by Layer	Yes	Identifies package, version and layer (OS/JAR/Node)

Tabela 17 - Results GQM Question 1

Question 2: Precision and Noise Quality (M4, M5)

Metric	Value	Observation
M4 - False Positive Rate	0% (0/2)	Alpine and hello-world without false positives
M5 - Version Precision	Good	Identifies fixed vs affected status

Tabela 18- Results GQM Question 2

Notable: Trivy provides fix availability information even for EOSL systems.

Question 3: Risk Assessment Consistency (M6, M7)

CVSS scores are reported from multiple authoritative sources (NVD, vendor advisories) with high NVD consonance. Additional contextual information includes fix status (fixed/affected), specific fixed versions, and links to CVE details.

Question 4: Performance and Integration (M8, M9, M10)

Metric	Value	Observation
M8 - Execution Time	Average: 0.5s	Extremely fast (excluding hello-world)
M9 - Database Size	Local (~300 MB)	Requires local storage
M10 - Output Quality	Excellent	Well-structured JSON

Tabela 19 - Results GQM Question 3

Performance by Complexity

Category	Package Range	Average Time
----------	---------------	--------------

Light	0-100	0.2s
Medium	100-500	0.5s
Heavy	500-1000	5.0s
Very Heavy	1000+	11.0s

Tabela 20 - Performance Trivy

5.2.7. Identified Strengths

1. Excellent detection of vulnerabilities in Java dependencies (JARs)
2. Comprehensive analysis of Node.js packages (package.json, node_modules)
3. Zero false positives in clean control images
4. Detailed vulnerability information including CVSS scores, fixed versions, and fix status
5. Extremely fast scans with average execution under 5 seconds
6. Local database enabling offline operation after initial download
7. Open-source and freely available
8. Multi-language support including Java, Node.js, Python, Go, Rust, and Ruby
9. Correct EOSL system analysis (unlike Docker Scout's empty JSON for old Debian)

5.2.8. Identified Limitations

1. Structural Blindness: Does not detect manually compiled software outside package managers
2. Application CVEs: Cannot detect application logic or configuration vulnerabilities
3. Metadata Dependency: Analysis limited to known package manager ecosystems
4. Potential gaps in CVE-to-package mapping (as demonstrated with SambaCry)
5. Local database requires approximately 300 MB storage
6. Periodic vulnerability database updates required for current coverage

5.2.9. Supplementary Data

Vulnerability Distribution by Severity

CRITICAL	~240	5.6%
HIGH	~907	21.3%
MEDIUM	~2,069	48.5%
LOW	~1,047	24.6%
TOTAL	4,263	100%

Tabela 21 - Vulnerability Distribution Trivy

Coverage by Package Type

Type	Analysed Images	Detection Success
OS Packages (Debian/Ubuntu)	10/10	100%
JAR (Java/Maven)	3/3	100%
Node.js (npm)	1/1	100%

Tabela 22 - Coverage by Package Trivy

5.3 Grype - Empirical Analysis

5.3.1 Consolidated Results (10 Images)

#	Image	Packages	Files	Total	CRIT	HIGH	MED	LOW	NEGL	Time (s)
1	alpine:latest	16	81	6	0	0	0	6	-	1.696
2	hello-world	0	1	0	0	0	0	0	-	35.330*
3	debian:stretch-slim	76	2713	209	23	46	20	46	74	1.808
4	bash (Shellshock)	132	7370	468	2	70	162	97	137	2.467
5	httpd:2.4.49	111	2857	398	34	134	113	29	84	4.165
6	spring-webmvc	190	6459	512	35	138	175	53	107	6.446
7	juice-shop	1132	2386	62	8	19	22	1	12	9.779
8	samba:4.6.3	330	15827	7353	1	223	5788	1341	-	16.857
9	solr (Log4Shell)	521	7246	918	32	211	421	124	130	10.331
10	confluence	1764	6415	889	24	219	467	155	24	19.678

Tabela 23 - Results Grype

Total Vulnerabilities: 10,815

Average Scan Time: ~8.1s (excluding hello-world with DB update)

- Includes vulnerability database update (~35s)

5.3.2 Target CVE Detection (Ground Truth)

CVE	Name	Image	Detected?	CVSS Min	CVSS Max	Observation
CVE-2014-6271	Shellshock	bash	YES	1.5	10.0	Classified as CRITICAL
CVE-2017-7494	SambaCry	samba	UNCERTAIN	1.2	10.0	Possible indirect detection
CVE-2021-44228	Log4Shell	solr	YES	1.2	10.0	Classified as CRITICAL
CVE-2022-22965	Spring4Shell	spring-webmvc	YES	1.5	10.0	Classified as CRITICAL

CVE-2022-22965	Spring4Shell	confluence	YES	1.9	10.0	Classified as CRITICAL
CVE-2022-26134	Confluence RCE	confluence	NO	-	-	Application vulnerability
CVE-2021-41773	Apache Path Traversal	httpd	NO	-	-	Application vulnerability
CVE-2021-42013	Apache Path Traversal	httpd	NO	-	-	Application vulnerability

Tabela 24 - CVE Detection Grype

Detection Metrics

- **Detection Rate:** 100% (3/3) of CVEs in dependencies†
- **True Positives:** 3-4 (Log4Shell, Spring4Shell, Shellshock, possibly SambaCry)
- **False Negatives:** 3 (Confluence RCE, 2× Apache Path Traversal)
- **False Positives:** Minimal (6 LOW in Alpine - debatable if FPs)

**Excluding application CVEs that are outside the scope of SBOM scanners*

5.3.3 Detailed Analysis by Category

Control Images (Baseline):

alpine:latest

- **Packages:** 16
- **Files:** 81
- **Vulnerabilities:** 6 (0 CRIT, 0 HIGH, 0 MED, 6 LOW)
- **Time:** 1.696s
- **CVSS Range:** 2.5 - 3.3

Analysis:

- Unlike Docker Scout and Trivy, Grype detected 6 LOW vulnerabilities
- Demonstrates greater **sensitivity** to theoretical vulnerabilities
- Question: Are these false positives or genuine vulnerabilities ignored by other tools?

hello-world:latest

- **Packages:** 0
- **Files:** 1 executable
- **Vulnerabilities:** 0
- **Time:** 35.330s (initial DB update cost)
- **Conclusion:** Correct behavior - empty image

EOSL Systems (End of Support Life):

debian:stretch-slim

- **OS:** Debian 9 (EOSL)
- **Packages:** 76
- **Files:** 2,713
- **Vulnerabilities:** 209 (23 CRIT, 46 HIGH, 20 MED, 46 LOW, 74 NEGL)
- **Time:** 1.808s
- **CVSS Range:** 1.2 - 10.0

bash (CVE-2014-6271) POTENTIAL SUCCESS

- **OS:** Debian 8 (EOSL)
- **Packages:** 132
- **Files:** 7,370
- **Vulnerabilities:** 468 (2 CRIT, 70 HIGH, 162 MED, 97 LOW, 137 NEGL)
- **Time:** 2.467s
- **CVSS Range:** 1.5 - 10.0

Analysis:

- Gripe reported 2 CRITICAL vulnerabilities with CVSS 10.0
- **Hypothesis:** One of these may be CVE-2014-6271 (Shellshock)
- Requires manual JSON verification to confirm
- **Differential vs. Scout/Trivy:** Gripe was not "blind" to compiled Bash

httpd:2.4.49

- **OS:** Debian 10 (EOSL)

- **Packages:** 111
- **Files:** 2,857
- **Vulnerabilities:** 398 (34 CRIT, 134 HIGH, 113 MED, 29 LOW, 84 NEGL, 4 UNKNOWN)
- **Time:** 4.165s
- **CVSS Range:** 1.2 - 10.0

Vulnerable Applications:

spring-webmvc:5.3.17 (Spring4Shell) - SUCCESS

- **OS:** Debian 11
- **Packages:** 190
- **Files:** 6,459
- **Vulnerabilities:** 512 (35 CRIT, 138 HIGH, 175 MED, 53 LOW, 107 NEGL, 4 UNKNOWN)
- **Time:** 6.446s
- **CVSS Range:** 1.5 - 10.0
- **CVE-2022-22965:** Detected

Observation: 35 CRITICAL vulnerabilities indicate deep Java ecosystem detection.

juice-shop:latest (OWASP)

- **OS:** Debian 12
- **Packages:** 1,132
- **Files:** 2,386
- **Vulnerabilities:** 62 (8 CRIT, 19 HIGH, 22 MED, 1 LOW, 12 NEGL)
- **Time:** 9.779s
- **CVSS Range:** 4.0 - 10.0

Analysis:

- CVSS minimum of 4.0 indicates no "light" vulnerabilities
- All flaws have at least MEDIUM severity
- Confirms that application libraries are inherently more dangerous than Alpine base packages

solr:8.11.0 (Log4Shell) - CRITICAL SUCCESS

- **OS:** Debian 8 (EOSL)
- **Packages:** 521
- **Files:** 7,246
- **Vulnerabilities:** **918** (32 CRIT, 211 HIGH, 421 MED, 124 LOW, 130 NEGL)
- **Time:** 10.331s
- **CVSS Range:** 1.2 - **10.0**
- **CVE-2021-44228:** Detected

Highlight: 32 CRITICAL vulnerabilities confirm deep JAR analysis.

confluence:7.13.6

- **OS:** Ubuntu 20.04
- **Packages:** 1,764
- **Files:** 6,415
- **Vulnerabilities:** **889** (24 CRIT, 219 HIGH, 467 MED, 155 LOW, 24 NEGL)
- **Time:** 19.678s
- **CVSS Range:** 1.9 - **10.0**
- **CVE-2022-22965 (Spring4Shell):** Detected
- **CVE-2022-26134 (Confluence RCE):** NOT detected (application)

Observation: Even with 1,764 packages, complete scan in < 20s.

Critical Case: Samba**samba:4.6.3 - MASSIVE OUTLIER**

- **OS:** Ubuntu 16.04 (EOSL)
- **Packages:** 330
- **Files:** **15,827** (highest number of files)
- **Vulnerabilities:** **7,353** (1 CRIT, 223 HIGH, 5788 MED, 1341 LOW)
- **Time:** 16.857s
- **CVSS Range:** 1.2 - **10.0**

CRITICAL DISCOVERY

The volume of **7,353 vulnerabilities** is a **massive outlier**.

Analysis:

- Packages:Vulnerabilities ratio = 1:22 (extremely high)
- 15,827 files analyzed (the largest)
- Gype does not deduplicate matches when the same vulnerable library appears in multiple files

Technical Explanation: If a library (e.g., libssl.so) is linked by 100 different binaries:

- Gype reports: **100 matches** of the same CVE
- Trivy reports: **1 unique** vulnerability

Conclusion:

- Not 7,353 unique vulnerabilities
- Approximately ~826 unique vulnerabilities propagated through 6,003 paths (according to Snyk analysis)
- **Explosion Ratio:** 1:7 (1 vulnerability - 7 matches)

5.3.4 Supplementary Data

Vulnerability Distribution by Severity

CRITICAL: ~159 (1.5%)

HIGH: ~1,280 (11.8%)

MEDIUM: ~7,368 (68.1%)

LOW: ~2,024 (18.7%)

NEGLIGIBLE: ~568 (outliers not counted)

TOTAL: 10,815

Observation: 68% are MEDIUM - indicates conservative classification profile.

Outlier Analysis

Metric	Without Samba	With Samba	Impact
Total Vulns	3,462	10,815	+212%
Average/Image	385	1,082	+181%

Tabela 25 - Supplementary Data Gype

Conclusion: Samba significantly distorts statistics.

5.3.5 GQM Metrics (Goal-Question-Metric)

Question 1: Detection Effectiveness (M1, M2, M3)

Metric	Value	Observation	Metric
M1 - True Positive Rate	100% (3/3 dependencies)	Shellshock, Log4Shell, Spring4Shell	M1 - True Positive Rate
M2 - False Negatives	3 application CVEs	Confluence RCE, 2× Apache PT	M2 - False Negatives
M3 - Detection by Layer	Partial	Reports package but does not explicitly indicate Docker layer	M3 - Detection by Layer

Tabela 26 - Results Grype Question 1

Highlight: Grype was the only one to **potentially** detect Shellshock (manually compiled).

Question 2: Precision and Noise Quality (M4, M5)

Metric	Value	Observation
M4 - False Positive Rate	Debatable	6 LOW in Alpine (other tools: 0)
M5 - Version Precision	Good	Correctly analyzes EOSL systems

Tabela 27 - Results Grype Question 2

Identified Problem:

- Match explosion in Samba (7,353) hinders triage
- May cause **alert fatigue** in old systems

Question 3: Risk Assessment Consistency (M6, M7)

Metric	Result
M6 - Severity Divergence	Raw NVD CVSS (no contextual adjustments)
M7 - NVD Consonance	Perfect - reports NVD values without modification

Tabela 28 - Results Grype Question 3

Question 4: Performance and Integration (M8, M9, M10)

Metric	Value	Observation
M8 - Execution Time	Average: 8.1s	Intermediate (Trivy: 0.5s, Snyk: 18s)
M9 - Database Size	Local (~500 MB)	Requires storage
M10 - Output Quality	Excellent	JSON, SARIF, CycloneDX, Table

Tabela 29 - Results Grype Question 4

Performance Analysis:

$$\text{Time (s)} \approx 0.011 \times \text{Packages} + 1.5$$

$$R^2 = 0.82$$

5.4 Snyc - Empirical Analysis

5.4.1 Consolidated Results (10 Images)

```
monteiro_20@LAPTOP-80QT8QD9:/mnt/c/Users/20joa/OneDrive/Ambiente de Trabalho/Lab/conteudo/snyk$ python3 vulnerabilities.py
```

Imagem	Deps	Unique	Crit	High	Med	Low	Paths
alpine	16	0	0	0	0	0	0
bash	Error	-	-	-	-	-	-
confluence	1604	451	13	110	244	84	1494
debian_slim	76	87	0	2	1	84	366
helloworld	0	0	0	0	0	0	0
httpd	110	223	19	52	39	113	812
juiceshop	983	72	5	26	29	12	135
log4j	496	548	84	256	182	26	1686
samba	324	826	0	13	419	394	6003
spring	157	253	23	55	52	123	859

Figure 41 - Total Vulnerabilities each image scanned

#	Image	Deps	Unique	CRIT	HIGH	MED	LOW	Paths	Time (s)
1	alpine	16	0	0	0	0	0	0	7.498
2	bash	Error	-	-	-	-	-	-	6.266
3	confluence	1604	451	13	119	244	84	1494	60.564
4	debian_slim	76	87	0	2	1	84	366	8.391
5	helloworld	0	0	0	0	0	0	0	6.205
6	httpd	118	223	19	61	39	113	815	18.214
7	juiceshop	583	223	5	51	29	12	135	21.942
8	log4j	496	548	84	256	182	26	1686	26.908
9	samba	324	826	0	419	394	6003	22.743	
10	spring	157	263	23	55	52	123	859	19.361

Tabela 30 - Results Snyc

Total Unique Vulnerabilities: ~2,621

Total Dependency Paths: ~11,358

Average Scan Time: ~18.8s (excluding alpine/hello-world)

5.4.2 Target CVE Detection (Ground Truth)

CVE	Name	Image	Detected?	CVSS	Exploit	Fixed In	Observation
CVE-2014-6271	Shellshock	bash	ERROR	-	-	-	Dockerfile not found
CVE-2017-7494	SambaCry	samba	UNCERTAIN	-	-	-	Possible indirect detection
CVE-2021-44228	Log4Shell	log4j	YES	10.0	Mature	2.15.0	Multiple paths

CVE-2022-22965	Spring4Shell	spring	YES	9.8-10.0	Mature	5.3.18	Multiple components
CVE-2022-22965	Spring4Shell	confluence	YES	9.8	Mature	5.3.18	Dependencies
CVE-2022-26134	Confluence RCE	confluence	NO	-	-	-	Application vulnerability
CVE-2021-41773	Apache Path Traversal	httpd	NO	-	-	-	Application vulnerability

Tabela 31 - CVE Detection Snyk

Detection Metrics

- **Detection Rate:** 66.7% (2/3 testable)
- **True Positives:** 2 (Log4Shell, Spring4Shell)
- **False Negatives:** 1+ (SambaCry, possibly more)
- **Errors:** 1 (bash - Dockerfile not found)

**Excluding bash (error) and application CVEs*

5.4.3 Snyk Features and Fixed In

Exploit Maturity

Critical Differentiator: Snyk indicates if public exploit code exists.

Image	Exploits	Fixes Avail
alpine	0	0
bash	0	0
confluence	1494	1335
debian_slim	366	6
helloworld	0	0
httpd	812	367
juiceshop	135	64
log4j	1686	1071
samba	6003	5598
spring	859	518

Figure 42 - Exploits and Fixed In

Impact for Prioritization:

CVSS 10.0 + Exploit "Mature" + No Fix = **URGENT**

CVSS 10.0 + No Exploit + Fix Available = **MANAGEABLE**

Fixed In (Fix Version)

Operational Advantage: Snyk provides the exact version that fixes the vulnerability.

Example (Log4Shell):

```
{  
  "CVE": "CVE-2021-44228",  
  "Package": "log4j-core",  
  "Current Version": "2.14.1",  
  "Fixed In": "2.15.0",  
  "Upgrade Path": "log4j-core@2.14.1 → 2.15.0"  
}
```

MTTR Reduction: Administrator doesn't need to investigate, already has the solution.

Dependency Paths

Differential vs. Other Tools: Snyk maps how the vulnerability entered the system.

Example:

App -> Spring Boot -> Spring Framework -> log4j-core@2.14.1 (VULNERABLE)

"Explosion" Metric:

- **Samba:** 826 unique vulnerabilities → 6003 paths (Ratio 1:7)
- **Juice-Shop:** 72 unique → 135 paths (Ratio 1:2)

Conclusion:

"The 'Vulnerable Dependency Paths' metric demonstrates that remediation complexity is greater than indicated by unique vulnerabilities alone, since the same flaw propagates through multiple paths in the dependency tree."

Snyk Priority Score

Adjusted CVSS: Snyk does not use only raw NVD CVSS.

Adjustment Factors:

1. Exploit Maturity (PoC available?)
2. EPSS (Exploit Prediction Scoring System)
3. Execution context (container vs. bare metal)
4. Prevalence in real attacks

Example:

- NVD CVSS: 10.0
- Snyk Score: 9.8 (adjusted by context)

5.4.4 Detailed Analysis by Category

Control Images (Baseline):

alpine:latest

- **Deps:** 16
- **Unique Vulnerabilities:** 0
- **Time:** 7.498s
- **CVSS Range:** N/A
- **Conclusion:** Clean image, no false positives

Performance Analysis:

- Fixed overhead of ~6-7s (API authentication + network)
- Even clean images have base cost

helloworld:latest

- **Deps:** 0
- **Vulnerabilities:** 0
- **Time:** 6.205s
- **Conclusion:** Confirms absence of false positives

EOSL and Vulnerable Systems:

debian:stretch-slim

- **OS:** Debian 9 (EOSL)
- **Deps:** 76
- **Unique Vulnerabilities:** 87
- **Severity:** 0 CRIT, 2 HIGH, 1 MED, 84 LOW
- **Paths:** 366
- **Time:** 8.391s
- **CVSS Range:** 2.0 - 9.8

Observation: Majority LOW due to EOSL system without fixes.

bash (CVE-2014-6271) - ERROR

- **Time:** 6.266s
- **Result:** Error: Dockerfile not found

Error Analysis:

- Snky expected a Dockerfile in the context
- Vulhub image did not provide accessible Dockerfile
- **Limitation:** Requires build context or registry analysis

httpd:2.4.49

- **Deps:** 118
- **Unique Vulnerabilities:** 223
- **Severity:** 19 CRIT, 61 HIGH, 39 MED, 113 LOW
- **Paths:** 815
- **Exploits:** 815 available
- **Fixes:** 507 available
- **Time:** 18.214s
- **CVSS Range:** 2.0 - 9.8

Vulnerable Applications:

juiceshop:latest

- **Deps:** 583
- **Unique Vulnerabilities:** 223 (SCA app layer)
- **Severity:** 5 CRIT, 51 HIGH, 29 MED, 12 LOW
- **Paths:** 135 (linear Node.js deps structure)
- **Time:** 21.942s
- **CVSS Range:** 2.3 - 10.0

Severity Analysis:

Log4j (Solr): 84 Critical, 256 High -> Superior immediate risk

Juice-Shop: 5 Critical (SCA app layer)

Highlight: The 5 critical ones come from **application layer** (SCA), validating Snyk use to detect what Gripe/Scout may miss.

log4j (solr:8.11.0) - CRITICAL SUCCESS

- **Deps:** 496
- **Unique Vulnerabilities:** 548
- **Severity:** 84 CRIT, 256 HIGH, 182 MED, 26 LOW
- **Paths:** 1686 (Java dependency explosion)
- **Exploits:** 1686 available
- **Time:** 26.908s
- **CVSS Range:** 1.2 - 10.0

Log4Shell Detected – SUCCESS

CVE	Package	CVSS	Exploit	Fixed In	Impact
CVE-2021-44228	log4j-core@2.14.1	10.0	Mature	2.15.0	Remote Code Execution
CVE-2021-45046	log4j-core@2.14.1	9.0	Mature	2.16.0	DoS / RCE

Tabela 32 - Log4Shell Detected

Observation: Despite having fewer dependencies than Juice-Shop (496 vs 583), took longer (26.9s vs 21.9s) due to **complexity of analyzing 84 CRIT and 1686 paths**.

spring-webmvc:5.3.17 - SUCCESS

- **Deps:** 157
- **Unique Vulnerabilities:** 263
- **Severity:** 23 CRIT, 55 HIGH, 52 MED, 123 LOW
- **Paths:** 859
- **Exploits:** 859 available
- **Time:** 19.361s
- **CVSS Range:** 2.3 - 10.0
- **CVE-2022-22965 (Spring4Shell):** Detected

Detection details:

Package	Version	CVE	CVSS	Fixed In
spring-beans	5.3.17	CVE-2022-22965	9.8	5.3.18
spring-webmvc	5.3.17	CVE-2022-22965	9.8	5.3.18

Tabela 33 - Detection Details Spring

confluence:7.13.6

- **Deps:** 1604 (highest)
- **Unique Vulnerabilities:** 451
- **Severity:** 13 CRIT, 119 HIGH, 244 MED, 84 LOW
- **Paths:** 1494
- **Time:** 60.564s (slowest)
- **CVSS Range:** 2.0 - 9.8

CVE Analysis:

- **CVE-2022-22965 (Spring4Shell):** Detected in dependencies
- **CVE-2022-26134 (Confluence RCE):** NOT detected (application OGNL)

samba:4.6.3 - "EXPLOSION" CASE

- **Deps:** 324
- **Unique Vulnerabilities:** 826
- **Severity:** 0 CRIT, 13 HIGH, 419 MED, 394 LOW
- **Paths:** 6003 (highest - ratio 1:7.3)
- **Time:** 22.743s
- **CVSS Range:** 2.0 - 8.1

Critical Observation:

- Ratio of **1:7.3** (each unique vuln → 7.3 paths)
- Demonstrates **remediation complexity** far exceeds unique count
- **CVE-2017-7494 (SambaCry):** Uncertain if detected

5.4.5 Performance Analysis (M8)

Fixed Cloud Overhead

Evidence:

Alpine: 7.498s (~0 vulns)

Bash: 6.266s (error)

Helloworld: 6.205s (~0 vulns)

Conclusion: Fixed overhead of ~6-7 seconds (API authentication + network).

Deps vs. Time Correlation

Evidence:

- Confluence (1604 deps) - 60.564s
- Debian Slim (76 deps) - 8.391s

Conclusion: The factor that most delays Snyk is the **size of the dependency tree**, not the number of vulnerabilities.

SCA Impact (Software Composition Analysis)

Scans with programming languages are slower:

- Juiceshop (21.942s) - Node.js (package-lock.json)
- Spring (19.361s) - Java (pom.xml)
- Log4j (26.908s) - Java (pom.xml) + calculation of 1686 paths

Detection Efficiency (Vulns/Time Ratio)

Image	Unique Vulns	Time	Vulns/s	Efficiency
Samba	826	22.743s	36 vulns/s	High
Log4j	548	26.908s	20 vulns/s	Medium

Tabela 34 - Performance Analysis Snyk

Conclusion: Samba scan was more "efficient" in vulns/time, but Log4j was more "dense" due to flaw severity (84 CRIT).

Performance Summary:

1. **Base latency:** 6-8s (Cloud-DB overhead)
2. **Scalability:** Linear with deps (not with vulns)
3. **Path Complexity:** High path density (e.g., Samba 6003) increases time
4. **Conclusion:** For fast CI/CD pipelines, Snyk is "slow" vs. Trivy/Grype, but extra time is justified by **analysis depth**.

5.4.6 Advanced Analysis: CVSS Vector String

Snyk does not "invent" a score; instead, it utilizes a globally standardized methodology called CVSS (Common Vulnerability Scoring System).

When explaining how Snyk calculates and uses CVSS in your report, you should focus on the fact that CVSS is a modular scoring system. Snyk does not just look at the "danger" of the code itself, but rather at how it can be attacked and what the consequences are if the attack succeeds.

Here is a technical and clear structure to explain this:

What is the "Vector String"?

It is the "formula" that defines the severity of the vulnerability. In your report, you can state that Snyk utilizes version 3.1 to ensure greater precision in assessing modern systems (cloud and containers).

Exploitability Metrics ("How easy is it?")

Snyk uses these metrics to determine if anyone on the internet can attack you or if it requires a specialist with physical access.

Attack Vector (AV): Defines if the attack is via Network, Adjacent, Local, or Physical.

Attack Complexity (AC): Whether the attacker needs "luck" or special conditions (High) or if it works every time (Low).

Privileges Required (PR): Whether the attacker needs to be an administrator, a common user, or no one (None).

User Interaction (UI): Whether someone needs to click a link for the exploit to work.

Impact Metrics ("How severe is it?")

This is where Snyk measures the damage to the famous security triangle (CIA Triad):

Confidentiality (C): Can the attacker read your data?

Integrity (I): Can the attacker modify or delete your files?

Availability (A): Can the attacker crash the server?

The "Scope" (S) – The CVSS v3.1

Differentiator This is a crucial technical point for your report. The Scope indicates whether a vulnerability in one component (e.g., a Docker container) can "escape" and affect the rest of the system (e.g., the host/physical machine).

Scope Unchanged (U): The damage is contained within the component.

Scope Changed (C): The vulnerability is much more dangerous because it can compromise the entire infrastructure.

```
monteiro_20@LAPTOP-89QT8QD9:/mnt/c/Users/20joa/OneDrive/Ambiente de Trabalho/Lab/conteudo/snyk$ python3 extract_minmax_cvss.py
```

Image	Min CVSS	Max CVSS
alpine	N/A	N/A
bash	N/A	N/A
confluence	2.3	9.8
debian_slim	2.4	9.8
helloworld	N/A	N/A
httpd	2.4	9.8
juiceshop	3.3	9.8
log4j	2.3	10.0
samba	2.4	9.8
spring	2.3	9.8

Figure 43 - Min and Max CVSS

5.4.7 Supplementary Data

Vulnerability Distribution by Severity

Unique Vulnerabilities:

CRITICAL: ~144 (5.5%)

HIGH: ~963 (36.7%)

MEDIUM: ~941 (35.9%)

LOW: ~573 (21.9%)

TOTAL: 2,621

Path Density:

Total Paths: 11,358

Unique: 2,621

Average Ratio: 1:4.3 (1 vuln - 4.3 paths)

Minimum vs. Maximum CVSS Analysis

Risk Amplitude:

Image	Min	Max	Gap	Interpretation
Log4j	2.3	10.0	7.7	Contains total RCE
Spring	2.3	9.8	7.5	Nearly total RCE
Samba	2.0	8.1	6.1	No 10.0 - less catastrophic
Juice-Shop	4.0	10.0	6.0	High minimum - all relevant
HTTPd	2.0	9.8	7.8	Nearly total RCE

Tabela 35 - Risk Amplitude Snyk

Conclusion:

"The presence of CVSS 10.0 values in images like Log4j and Spring validates the need for deep scanning tools, since these critical flaws, although minority in number, represent a risk of total infrastructure compromise."

Remediation Cost (Effort)

Effort Metrics:

Image	Unique Vulns	Paths	Ratio	Complexity
Samba	826	6003	1:7.3	VERY HIGH
Log4j	548	1686	1:3.1	HIGH
Confluence	451	1494	1:3.3	HIGH
Spring	263	859	1:3.3	MEDIUM
Juice-Shop	223	135	1:0.6	LOW

Tabela 36 - Effort Snyk

Conclusion:

"In the Samba scenario, the tool identified 826 unique vulnerabilities, which however branch into 6003 dependency paths. This discrepancy is one of the main factors of ambiguity in security assessment."

5.4.8 GQM

Question 1: Detection Effectiveness (M1, M2, M3)

Metric	Value	Observation
M1 - True Positive Rate	66.7% (2/3 testable)	Log4Shell and Spring4Shell successfully detected
M2 - False Negatives	3 CVEs	SambaCry (uncertain), Confluence RCE, Apache PT
M2.1 - False Negative Rate by Severity	CRITICAL: 0/2 (0%), Application-level: 3/3 (100%)	Failures only in application-level CVEs
M3 - Detection by Layer	Excellent	Identifies package, version, layer (OS/JAR/Node) AND full path

Tabela 37 - Results Snyk Question 1

Question 2: Precision and Noise Quality (M4, M5)

Metric	Value	Observation
M4 - False Positive Rate	0% (0/2)	Alpine and hello-world without false positives
M5 - Version Precision (Backports)	Excellent	Recognizes backports via Debian Security Advisories

Tabela 38 - Results Snyk Question 2

Question 3: Risk Assessment Consistency (M6, M7)

Metric	Result
M6 - Severity Divergence	Divergence Index: 0.31 (low)

M7 - Consonance with NVD	Medium (75%) - uses Snyk DB + NVD
M11 - Majority Consensus	89% (8/9 testable images)

Tabela 39 - Results Snyk Question 3

Question 4: Performance and Integration (M8, M9, M10, M12)

Metric	Value	Observation
M8 - Execution Time (Cold Start)	Average: 18.8s	Slower than Grype (8.1s) and Trivy (0.5s)
M8 - Execution Time (Warm Start)	Average: 16.2s	Cache reduces ~2.6s
M9 – Database Size	0 MB local	Cloud-based (requires internet)
M10 - Output Quality	Excellent	JSON + rich metadata (exploit, fixed in, paths)
M12 - CI/CD Integration Ease	Medium (3/5)	Requires authentication and internet

Tabela 40 - Results Snyk Question 4

6. Empirical Results

6.1 Question 1: Detection Effectiveness (M1, M2, M3)

6.1.1 Validation of the "Fallacy of Total Coverage" - Manually Compiled Software

Scanner	Total Vulns (bash)	Shellshock Detected?	Observation
Trivy	258	NO	Bash compiled via <code>./configure && make</code>
Snyk	N/A (Error)	N/A	Dockerfile not found
Grype	468	NO	Detected base OS vulns, not Bash
Docker Scout	0	NO	Empty JSON (Debian 8 EOSL)

Tabela 41 - Empirical Results **Shellshock** Question 1

Analysis: None of the four tools were able to detect CVE-2014-6271 (Shellshock), despite the image explicitly containing vulnerable Bash 4.3. This limitation, called "structural blindness" (Plate et al., 2015), occurs because the software was manually compiled from source code:

```
RUN wget -qO- <https://ftp.gnu.org/gnu/bash/bash-4.3.tar.gz> | tar zx
```

```
RUN ./configure --prefix=/usr/local/bash-4.3.0 && make && make install
```

Conclusion M2: False Negative Rate = 100% for manually compiled software. This finding empirically validates the "Fallacy of Total Coverage" hypothesis (Section 3.2).

6.1.2 Detection of CVEs in Java Libraries (True Positives)

CVE	Name	Type	Trivy	Snyk	Grype	Docker Scout
CVE-2014-6271	Shellshock	Manual Compilation	Not detected	N/A	Not detected	Not detected
CVE-2017-7494	SambaCry	Library	Not detected	Not detected	Not detected	Not detected
CVE-2021-44228	Log4Shell	Java Library	Detected	Detected	Detected	Detected
CVE-2022-22965	Spring4Shell	Java Library	Detected	Detected	Detected	Detected
CVE-2022-26134	Confluence RCE	Application	Not detected	Not detected	Not detected	Not detected
CVE-2021-41773	Apache Path Traversal	Application	Not detected	Not detected	Not detected	Not detected

Tabela 42 - CVE Detection Java Libraries

Global Detection Rate (M1): 2/6 target CVEs = **33.3%**

Conclusion: There is unanimity in detecting Log4Shell and Spring4Shell (well-documented Java libraries) and in the non-detection of application vulnerabilities and manually compiled software.

6.1.3 Infrastructure Vulnerabilities (M3 - Detection by Layer)

Image	Trivy	Snyk	Grype	Docker Scout
alpine:latest	0	0	6	0
debian:stretch-slim	68	87	209	0
Maximum difference	-	-	-	209 vs 0

Tabela 43 - Detection Comparison in Base Images

Critical Analysis: Docker Scout presents anomalous behavior, reporting **zero vulnerabilities** in discontinued Debian systems, while the remaining tools identify between 68 and 209 CVEs.

6.2 Question 2: Precision and Noise Quality (M4, M5)

6.2.1 Clean Image Baseline - False Positives (M4)

Image	Trivy	Snyk	Grype	Docker Scout	Consensus
hello-world:latest	0	0	0	0	Unanimous
alpine:latest	0	0	6 (LOW)	0	Divergence

Tabela 44 - Control Image Analysis

Analysis: The hello-world:latest image, consisting only of a static binary without an operating system, serves as an ideal negative control. All tools agree on the absence of vulnerabilities, **validating the basic precision of scanners** (M4 = 0 false positives in pure negative controls).

The divergence in Alpine (Grype reports 6 LOW vulnerabilities) illustrates differences in policies for including low-severity CVEs.

6.2.2 EOSL System Management - Technical Lag (M5)

Image	OS	EOSL	Trivy	Snyk	Grype	Docker Scout
debian:stretch-slim	Debian 9.13	2022	68	87	209	0
bash/Shellshock	Debian 8.10	2020	258	N/A	468	0
vulhub/httpd	Debian 10.10	2024	260	223	398	0
vulhub/samba	Ubuntu 16.04	2021	1579	826	3045	848

Tabela 45 - Behavior in Discontinued Systems

Critical Discovery: Docker Scout presents **inconsistent** behavior between EOSL distributions:

- **Debian 8/9/10:** Empty JSON (0 reported vulnerabilities)
- **Ubuntu 16.04:** 848 vulnerabilities correctly identified

This inconsistency **validates the "Hidden Debt of Technical Lag" dimension** (Section 3.1), security technical debt remains invisible depending on the tool used.

Scanner	Distinguishes Fixed/Affected?	Example
Trivy	Yes	219 fixed, 39 affected
Snyk	Yes	"Fixed In" field with specific version
Grype	Yes	By severity
Docker Scout	Yes	Fixed version indicated

Tabela 46 - Fixed vs. Affected Distinction (M5)

6.3 Question 3: Risk Assessment Consistency (M6, M7)

6.3.1 Severity Divergence (M6)

Image	Scanner	CRITICAL	HIGH	MEDIUM	LOW	TOTAL
vulhub/solr (Log4Shell)	Trivy	121	264	258	77	747
	Snyk	84	256	182	26	548
	Grype	32	211	421	124	918
	Docker Scout	31	54	47	8	140
vulhub/samba	Trivy	1	68	987	523	1579
	Snyk	0	13	419	394	826
	Grype	1	223	5788	1341	7353*
	Docker Scout	~10	~80	~400	~350	848

- Tabela 47 - Severity Classification by Scanner Note: The Grype value includes duplication by dependency paths.

Analysis: The vulhub/solr:8.11.0 image exemplifies the "**risk assessment ambiguity**" (Section 3.3): the same system is classified with 747 vulnerabilities (Trivy), 548 (Snyk), 918 (Grype) or 140 (Docker Scout).

M6 Calculation: Variation of **6.5x** between maximum (918) and minimum (140), compromising metric comparability between organizations.

6.3.2 CVSS Amplitude (M7 - NVD Consonance)

Image	Snyk Min	Snyk Max	Grype Min	Grype Max
Log4j/Solr	2.3	10.0	1.2	10.0
Confluence	2.3	9.8	1.9	10.0
Samba	2.4	9.8	1.2	10.0
Spring	2.3	9.8	1.5	10.0

Tabela 48 - CVSS Amplitude by Tool

Observation: Grype tends to report the absolute maximum value (10.0) from NVD without contextual adjustments, while Snyk applies its own "Snyk Score" that considers exploit maturity. This divergence **empirically validates the "Ambiguity and Inconsistency" dimension** (Section 3.3).

6.4 Question 4: Operational Performance (M8, M10)

6.4.1 Execution Time (M8)

Methodological Note: The execution times presented were obtained in a specific laboratory environment and may vary significantly depending on the characteristics of the hardware used, network bandwidth, system load, and cache state of vulnerability databases.

Image	Trivy	Snyk	Grype	Docker Scout
alpine:latest	0.236	7.498	1.696	3.907
hello-world:latest	11.985*	6.205	35.330*	2.280
debian:stretch-slim	1.378	8.391	1.808	6.947
bash/Shellshock	0.175	6.266	2.467	1.935
vulhub/httpd	0.123	18.214	4.165	10.653
vulhub/spring-webmvc	0.298	19.361	6.446	18.390
juice-shop:latest	10.681	21.942	9.779	25.283
vulhub/samba	6.710	22.743	16.857	5.703
vulhub/solr (Log4j)	0.659	26.908	10.331	2.253
vulhub/confluence	11.641	60.564	19.678	2.849
Average	4.39s	19.81s	10.86s	8.02s

Tabela 49 - Execution Time per Image (seconds)

- Includes database download/update time.

Scanner	Architecture	Base Latency	Scale Factor
Trivy	Local + Remote DB	~0.2s	Linear with packages
Snyk	Cloud-native (SaaS)	~6-7s	Linear with dependencies
Grype	Local + Remote DB	~1.7s	Linear with files
Docker Scout	SBOM + Cloud	~2.3s	Variable

Tabela 50 - Performance Analysis by Architecture

7. Synthesis and Hypothesis Validation

7.1 Mapping Results - Problem Dimensions

Dimension	Hypothesis	Empirical Result	Validation
2.1 Technical Lag	Scanners fail in EOSL systems	Docker Scout reports 0 vulns in Debian EOSL	Validated
2.2 Structural Blindness	Compiled software is invisible	0% Shellshock detection (4 tools)	Validated
2.3 Risk Ambiguity	Divergence between tools	Variation of 6.5x in the same image	Validated

Tabela 51 - Mapping Results

7.2 Global Evaluation Matrix

Numerical scale from 1 to 5, where 5 is the maximum score.

Criterion	Trivy	Snyk	Grype	Docker Scout
Package Manager Detection (M1)	5	4	5	3
Manual Compilation Detection (M2)	1	1	1	1
Java/Node.js Coverage (M1)	5	5	5	4
EOSL Systems (M5)	4	4	5	2
Low False Positives (M4)	5	5	4	5
CVSS Consistency (M6)	4	3	4	3
Performance (M8)	5	2	4	3
Output Quality (M10)	5	4	5	3

Tabela 52 - Global Evaluation Matrix

7.3 Recommendation by Use Case

Scenario	Recommended Tool	Justification (Metrics)
Fast CI/CD Pipeline	Trivy	M8: Best performance (<5s average)
Enterprise Risk Management	Snyk	M7: Exploit maturity + remediation paths
Complete Audit	Grype	M1: Greater coverage, including LOW
Native Docker Environment	Docker Scout	M10: Native integration with Docker Hub
Legacy/EOSL Systems	Grype or Trivy	M5: Better support for old distributions

Tabela 53 - Recommendation by Use Case

8. Conclusion

This empirical study evaluated four Docker container vulnerability scanners, Trivy, Grype, Snyk, and Docker Scout, using a Goal-Question-Metric (GQM) framework across ten Docker images with distinct risk profiles. The experimental results validated all three critical dimensions identified in the theoretical framework.

The Structural Blindness hypothesis (Section 3.2) was confirmed through a 0% detection rate for CVE-2014-6271 (Shellshock) across all tools, demonstrating that manually compiled software remains invisible to metadata-based scanning. The Technical Lag dimension (Section 3.1) was validated by Docker Scout's inconsistent EOSL handling, reporting zero vulnerabilities in Debian 8/9/10 while correctly detecting 848 CVEs in Ubuntu 16.04. The Risk Ambiguity dimension (Section 3.3) was quantified with a 6.5x variation in vulnerability counts for identical images, ranging from 140 to 918 CVEs depending on the tool.

Regarding the GQM research questions: detection effectiveness (Q1) yielded a 33.3% True Positive Rate for target CVEs, with unanimous success only for well-documented Java vulnerabilities (Log4Shell, Spring4Shell); precision analysis (Q2) confirmed zero false positives in negative controls; risk assessment (Q3) revealed significant inter-tool variance requiring organizational normalization policies; and performance evaluation (Q4) identified Trivy as the fastest tool (4.39s average) compared to Snyk (19.81s).

These findings demonstrate that while current scanners provide valuable insights for package-managed software, they exhibit significant blind spots. Organizations should adopt multi-tool strategies, implement supplementary controls such as SBOM validation and runtime monitoring, and verify scanner compatibility with legacy distributions. The validation of all theoretical framework dimensions underscores the need for comprehensive security strategies that extend beyond automated scanning alone.

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